

Jordan Separation, Cauchy Theory, and Stokes Calculus

on the Surcomplex Plane

Topologies, admissible chains, polynomial and deformed Stokes,
multiscale contours, and radius-free residues

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Abstract

A counterpart of a complex-analytic theorem over $\mathbf{SC} = \mathbf{No}[z]$ must specify three things before it has a definite meaning: a topology, a regularity algebra for the integrand, and a category of chains with an integration pairing. The full fine topology admits no nonconstant continuous ordinary real-parameter paths; a fixed Hahn workspace has a different, nondiscrete but totally separated intrinsic topology. Neither supports the literal ordinary Jordan-curve picture.

We separate the question into four theories and prove each in its own category. First, two Jordan separation theorems: a deliberately coarse one in the standard-part topology, whose separator is the whole boundary monad rather than a curve, and the established semialgebraic Jordan theorem over real closed fields, which we import rather than prove. Second, three chain categories carrying an exact generalized Stokes theorem: polynomial simplices over any characteristic-zero field, Hahn differential forms on ordinary manifolds with the cohomology comparison $H^q(\Omega_H^\bullet(M)) \simeq H_{\text{dR}}^q(M; \mathbb{C})((t^\Gamma))$, and infinitesimally deformed chains with a support-controlled Taylor pullback. Third, Cauchy theory as a period theory: an exact endpoint-transport formula, integer winding at displaced centers, a Cauchy formula on deformed Jordan contours, and moving-pole residues. Fourth, a positive-monomial rescaling theorem that converts radius-free Hahn analytic germs into polynomial-coefficient families, and with it microscopic circle and torus realizations of the perturbation residue, including a determinant-transformed realization for an arbitrary isolated complete intersection.

A rational contour pairing built from semialgebraic winding numbers is characterized by a uniqueness property and compared with Hahn integration on explicitly verified overlaps. Root-of-unity sampling is analyzed on both sides: a coefficientwise recovery theorem under an explicit decay hypothesis, and an exact example showing that the hypothesis cannot be dropped. Berkovich, tropical, perfectoid, definable, tame, and resurgent integration theories are compared without identifying their distinct targets.

Status. Established results, statements reported in the two supplied manuscripts, and deductions proved here are distinguished throughout. This is not a claim of a unique accepted foundation for surcomplex analysis, an exhaustive priority determination, or a formally verified proof. Symbolic checks validate the displayed finite examples and are not machine verification of the general proofs. All sums and mathematical workspaces used in the constructions are set-sized. The literature search was conducted on September 21, 2026, and was targeted rather than exhaustive; no named published conjecture is claimed to be solved.

Contents

1	The question, the sources, and the principal conclusions	1
1.1	Three theorems, three kinds of structure	1
1.2	What the supplied manuscripts provide	1
1.3	Established external counterparts	2
1.4	A map of the answers	2
1.5	Conventions, and one name for one concept	3
1.6	What is merged here, and from where	3
2	Workspaces, supports, and the four coefficient rings	4
2.1	A set-sized workspace	4
2.2	Strong summability is algebra, not a hidden limit	4
2.3	The four coefficient rings, named once and never interchanged	5
2.4	The perturbation residue, and why it is not a contour integral	6
3	Three topologies, and why the naive constructions fail	7
3.1	The full fine topology and its restriction	7
3.2	The intrinsic topology of a fixed Hahn workspace	8
3.3	The standard-part topology	8
3.4	Unrestricted fundamental theorems are impossible	8
3.5	Finite meshes cannot resolve infinitesimal tolerances	9
3.6	A contour avoiding every pole need not have an admissible pullback	10
4	Jordan separation: what survives, and in which sense	10
4.1	A thick-wall theorem in the standard-part topology	10
4.2	The semialgebraic replacement	11
4.3	An integer-valued index without a contour integral	12
4.4	What Jordan separation does not supply	13
5	A completely algebraic Stokes theorem for polynomial chains	13
5.1	Polynomial integration on a simplex	13
6	A Hahn de Rham complex and coefficientwise Stokes	14
6.1	The differential graded algebra of globally supported Hahn forms	14
6.2	A support-preserving Poincaré lemma	15
6.3	Cohomology and periods	16
6.4	Green, divergence, and area	17
7	Infinitesimally deformed chains, homotopy, and estimates	17
7.1	Admissible deformations and the Taylor-jet pullback	17
7.2	Affine scales and polynomial compatibility	20
7.3	The homotopy formula	20
7.4	An order-valued parameter estimate	21
8	Cauchy theory as a period theory	21
8.1	Coherent holomorphy and the vanishing theorem	22
8.2	Exact transport of an open contour	22
8.3	Winding at a displaced center, and the Cauchy formula	24
8.4	Moving poles and actual zero clusters	25
9	Affine scales, microscopic contours, and compatible charts	26
9.1	A precise compatibility criterion	26
9.2	Macroscopic and microscopic winding resolve different information	27

10 Radius-free germs become coherent after microscopic rescaling	27
10.1 The missing common-radius problem	28
10.2 Microscopic Cauchy formulae over $\mathcal{A}_n$	29
10.3 Protecting radii and one-variable residues	29
11 Microscopic tori and general complete-intersection residues	32
11.1 Coordinate-power leading systems	32
11.2 Arbitrary isolated analytic complete intersections	33
11.3 Relation to the finite-algebra conclusions of the supplied source	34
11.4 Scope: what a general isolated complete intersection still requires	35
12 An algebraic contour pairing for rational differentials	38
12.1 Definition from definable winding	38
12.2 Residues are the complete rational obstruction	39
12.3 Agreement with Hahn integration on a verified overlap	40
12.4 What this pairing does not determine	41
13 Root-of-unity sampling: a positive theorem and its indispensable hypothesis	41
13.1 What the classical construction assumes	41
13.2 A rank-independent sufficient condition for a valuation limit	42
13.3 Coefficientwise recovery always has a different meaning	42
13.4 Why the hypothesis cannot be dropped	43
14 Other established approaches and their exact scope	44
14.1 Semialgebraic chains, Stokes, and the target of integration	44
14.2 Berkovich geometry changes the space, not just the topology	45
14.3 Tropical differential forms and Stokes on Berkovich spaces	45
14.4 Perfectoid analytic cycles: an actual Stokes theorem, but a different base	45
14.5 Surreal antiderivatives, resurgent extensions, and nonstandard bridges	46
14.6 Comparison of the integration strategies	46
15 A usable procedure, and research directions with exact missing steps	47
15.1 Choose the object before choosing the formula	47
15.2 A compact decision table	48
15.3 Research problems sharpened by these results	48
16 Conclusion	49
A Support certificates and logical dependencies	50
B Why finite-parameter testing proves residue transformation	51
B.1 Finite dependency	51
B.2 Why a formal matrix causes no convergence gap	52
C What is claimed, what is imported, and what is not proved	52
C.1 Consolidated status statement	52
C.2 Per-theorem non-claims, preserved	53
D Notation	54
References	55

1 The question, the sources, and the principal conclusions

1.1 Three theorems, three kinds of structure

The three classical theorems in the title concern different structures. The Jordan curve theorem describes separation by a topological embedding of a circle. Cauchy's integral theorem describes the vanishing of periods of holomorphic differential forms under a homological hypothesis. Generalized Stokes relates a differential on forms to a boundary on chains. They cooperate in ordinary complex analysis but are not logically interchangeable: even over $\mathbb{C}$, a Jordan curve need not be rectifiable, a contour need not be simple for Cauchy's theorem to apply, and being closed does not make a cycle null-homologous in a punctured domain. One should therefore not try to extend the three theorems to $\mathbf{SC} = \mathbf{No}[i]$ by a single replacement of $\mathbb{C}$ by a larger field.

Real closedness of $\mathbf{No}$ supplies algebraic closedness of $\mathbf{SC}$, but it does not select a topology, a function algebra, or a chain category. The useful answer is not one unrestricted transplantation but a collection of precisely compatible counterparts. The classical starting point is the usual contour, Cauchy, and differential-form theory [3, 26, 19].

The positive conclusions are these. A genuine Jordan analogue is available in *semialgebraic topology* over every real closed Hahn workspace. A second, intentionally coarse Jordan statement is available after passing to standard parts. An exact $\mathbf{SC}$ -valued Stokes calculus is available in three separate chain categories: polynomial simplices, Hahn-coherent chains over an ordinary base, and infinitesimal deformations of the latter. Cauchy theory becomes a closed-form period theory rather than a consequence of fine connectedness. Finally, positive monomial rescaling makes radius-free analytic germs accessible to contour methods, and rational differentials admit a contour pairing based on algebraic winding numbers even when an ordinary leading contour meets the reduction of a pole. These answers coexist; they are not interchangeable.

1.2 What the supplied manuscripts provide

We refer to *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory* as the *analysis manuscript* [1]. It distinguishes fine-local power-series calculus from the algebra

$$\mathcal{H}(U) = \mathcal{O}(U)((t^\Gamma))$$

of well-ordered Hahn families of ordinary holomorphic functions on one common ordinary domain U . Its section on contour functionals defines integration coefficientwise on ordinary paths and proves Cauchy, Morera, primitive, and integral-estimate theorems; later sections prove moving-pole residue formulas, permit coherent affine changes of scale, and construct a global fine-analytic logarithm. These are existing results within the supplied framework.

The second source, *Infinitesimal Analytic Geometry over the Surcomplex Numbers: Radius-Free Hahn Germs, a Nullstellensatz, and Conservation of Singular Zeros*, is the *geometry manuscript* [2]. It replaces common-domain coefficient functions by

$$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma)),$$

where each coefficient is a convergent complex germ but the coefficients need not share a radius. For isolated complete intersections it defines a perturbation residue by a strongly summable series of ordinary local residues, and it explicitly does *not* define that functional by a surcomplex contour integral. The gap between a well-defined residue and a representing contour is one principal topic below.

Neither manuscript establishes a Jordan theorem for arbitrary fine-topological curves, nor generalized Stokes for arbitrary fine-smooth functions on arbitrary surcomplex domains. We do not silently attribute either assertion to them. We retain their distinction between ordinary coefficient regularity, strong summability, and fine differentiation. This article is not an independent audit of every theorem in either supplied manuscript.

1.3 Established external counterparts

There are relevant results beyond the supplied sources. Berarducci and Otero develop definable intersection numbers, winding numbers, and Jordan–Brouwer separation over o-minimal expansions of real closed fields [5, 6, 7]; Woerheide’s o-minimal homology is the other route to definable Jordan–Brouwer [30]. Eisermann gives an algebraic winding-number construction using Cauchy indices and Sturm chains over real closed fields [15], with corrections and boundary cases treated by Perrucci and Roy [25]. Ohmoto and Shiota give C^1 semialgebraic triangulations and a Stokes formula, and explicitly allow a larger saturated target field [23]; Maříková and Shiota measure definable sets with values in an ordered semiring [21]; Kaiser develops tame semialgebraic integration over real closed fields with Archimedean value group [18]. Chambert-Loir and Ducros construct a Stokes theory for real forms on Berkovich spaces [12], whose underlying topology is developed by Baker–Rumely and Temkin [8, 27]. Mihara constructs analytic singular homology and an integration theory over cosimplicial perfectoid spaces [20]. Cherry and Diamond develop Schnirelman-type and p -adic line integrals [11, 14]; a contemporary use of such averages appears in [16]. Ducros, Hrushovski, and Loeser construct a comparison between complex and non-Archimedean integrals [13]. Costin and Ehrlich study surreal extension and integration of distinguished classes of real functions [10].

These works answer substantial parts of the question in their own categories. A theorem about real-valued tropical forms is not already a theorem about K -valued holomorphic contour integrals. A definably connected interval is not an ordinarily connected interval. An integral valued in a saturated extension or an ordered semiring is not automatically a canonical element of a previously fixed Hahn field.

1.4 A map of the answers

Question	Valid counterpart	Essential qualification
Jordan separation	Definable circle in a real closed plane	Definably connected components, not ordinary ones
Coarse Jordan separation	Standard-part preimage of an ordinary Jordan curve	A thick wall in a non-Hausdorff topology
Generalized Stokes, algebraic	Polynomial simplices over any characteristic-zero field	A finite factorial identity, not a measure
Generalized Stokes, analytic	Hahn forms on ordinary or infinitesimally deformed chains	Support-controlled pullback and integration
Cauchy vanishing	Closed Hahn-holomorphic 1-form on an admissible null-homologous contour	Coherent coefficient data, or a specified algebraic pairing
Contour deformation	Closed-form homotopy invariance; exact endpoint corrections	The whole homotopy must be admissible and avoid singular coefficients
Radius-free residues	Microscopic circle, or torus after positive monomial rescaling	A determinant transform for a general complete intersection

Question	Valid counterpart	Essential qualification
Arbitrary rational loops	Residue functional weighted by algebraic winding	A definition with compatibility theorems, not a Riemann limit
Sampling a circle	Coefficientwise root-of-unity recovery, or a verified valuation-tail criterion	Coefficientwise, valuation, and fine convergence are not interchangeable

The affirmative rows are theorems in specified categories, not approximations to a nonexistent unrestricted fine-topological contour theory.

1.5 Conventions, and one name for one concept

Three of the source manuscripts merged here use three different names for one and the same topology, and several symbols are used inconsistently across them. We fix the following conventions and use them without exception.

1. **The standard-part topology.** The initial topology of the standard-part map on the finite elements — whose open sets are exactly the preimages of ordinary open subsets of $\mathbb{C}$, whose fibres are indiscrete, and whose Hausdorff quotient is $\mathbb{C}$ — is called the *standard-part topology* throughout. It appears in the literature merged here under the names *residue topology*, *reduction topology*, and *shadow topology*; these are synonyms for one and the same topology, and are not used again. The descriptive name is preferred here because “residue” is already committed to the residue functionals of [sections 10](#) and [11](#) and “shadow” is used in the analysis manuscript for the ordinary shadow of a zero.
2. **One symbol per concept.** We write st for the standard-part (reduction) map $K^\circ \rightarrow \mathbb{C}$ and never red ; $\mathfrak{m}_K$ for the infinitesimals of the workspace and $\mathfrak{m}$ only for the class of all infinitesimals of $\mathbf{SC}$; $U^\sharp = \text{st}^{-1}(U)$ for the thickening of an ordinary set; v for the valuation of a field element and v_H for the least exponent of a Hahn coefficient family; wind for ordinary complex winding and wind_{sa} for semialgebraic (definable, algebraic) winding, never Ind or w_F ; res for an ordinary local residue and Res_F for the perturbation residue of [\(2.5\)](#) and [\(2.4\)](#).
3. **Scales.** Rescaling exponents are always written $\rho_j \in \Gamma_{>0}$ and the corresponding radii $r_j = t^{\rho_j}$. The sources use ρ , β , and s for the exponent and r , ρ , and r for the radius; only the first convention appears below.
4. **Multiplicities.** Three senses are kept apart and never conflated: the order $m \geq 1$ of an ordinary one-variable germ at zero; the global colength $\mu = \dim_{\mathbb{C}} \mathbb{C}\{z\}/(f_1, \dots, f_n)$ of an isolated complete intersection; and the local multiplicity $e_a = \dim_K B_a$ at an actual displaced zero, with $\sum_a e_a = \mu$. The coordinate exponents in a coordinate-power system are written d_j .
5. **Every theorem names its ring.** Four coefficient rings occur below under nearly identical notation, and no theorem is transferred between them. They are tabulated in [section 2.3](#).

1.6 What is merged here, and from where

This article merges three manuscripts of September 21, 2026 that share roughly two thirds of their content: *Stokes Calculus*, *Stokes Integration*, and *Stokes Formulae on the Surcomplex Plane*. The shared material — the fine-topology no-go results, the standard-part Jordan theorem, the imported semialgebraic Jordan theorem, the Hahn form complex and its cohomology, deformed Stokes, the

endpoint-displacement formula, the rescaling lemma, and the one-variable microscopic circle — is stated once.

The differences are real and sectional. The polynomial-chain theory of [section 5](#), the support-preserving Poincaré lemma [proposition 6.2](#), the order-valued estimate [proposition 7.6](#), the sampling theory of [section 13](#), and the scope discussion of [section 11.4](#) come from the second manuscript alone. The uniqueness characterization of the algebraic pairing [theorem 12.2](#), the two compatibility results [propositions 9.1](#) and [12.4](#), the quantitative finite-partition obstruction [proposition 3.5](#), and above all the unconditional general contour representation [lemma 11.3](#) and [theorem 11.4](#) come from the first. The separately numbered winding-stability theorem [theorem 8.4](#), the numbered admissible-deformation definition [definition 7.1](#), the independent separating-torus route of [theorem 11.5](#), and the finite-parameter appendix [section B](#) come from the third.

Two of these overlap in a way that is a genuine disagreement about status rather than about arithmetic, and one is a disagreement about emphasis. Both are resolved in the open in [section 11.4](#) and [section 13](#) respectively, rather than averaged away.

2 Workspaces, supports, and the four coefficient rings

2.1 A set-sized workspace

Write $\mathbf{SC} = \mathbf{No}[i]$ and use the surreal normal-form monomials $t^\gamma = \omega^{-\gamma}$. Let $\Gamma \subset \mathbf{No}$ be a nonzero divisible set-sized ordered additive subgroup and put

$$R_\Gamma = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)) = R_\Gamma[i], \quad t^\gamma = \omega^{-\gamma}. \quad (2.1)$$

An element is a series whose exponent support is well ordered. Its valuation $v(a)$ is its least exponent, with $v(0) = +\infty$. The Hahn-field theorem makes R_Γ real closed and K algebraically closed; the algebraic-closure statement is Corollary 4 of Poonen [\[24\]](#). The normal-form interpretation places both fields inside $\mathbf{SC}$; the surreal background and restricted analytic evaluation are discussed by van den Dries and Ehrlich [\[29\]](#). Put

$$K^\circ = \{a : v(a) \geq 0\}, \quad \mathfrak{m}_K = \{a : v(a) > 0\}, \quad \text{st} : K^\circ \longrightarrow \mathbb{C},$$

where st extracts the coefficient of t^0 . Every finite element is uniquely $a = c + \varepsilon$ with $c \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$. For ordinary $U \subset \mathbb{C}$ write $U^\# = \text{st}^{-1}(U)$. The modulus $|x + iy| = \sqrt{x^2 + y^2}$ is valued in R_Γ , not in $\mathbb{R}$; it satisfies the algebraic triangle inequality and is not a real-valued non-Archimedean norm.

Enlargement rule. Every finite collection of surcomplex points, and every set-sized coefficient presentation used below, lies in a workspace of this kind, obtained by adjoining the groups generated by the relevant normal-form supports and taking the divisible hull. *Its limit.* This does not authorize a sum over a proper class, nor a proper-class polynomial ring, nor does it imply that every rescaling required later already lies in a previously chosen Γ . Differentiation in a variable z holds every element of K fixed; it is not a derivation on $\mathbf{No}$ regarded as a transseries field. Full-class neighborhood statements may be interpreted in a class theory such as NBG, while all rings, supports, and parameter manifolds used in the integration constructions are set-sized.

2.2 Strong summability is algebra, not a hidden limit

Definition 2.1 (Strong summability). A family of Hahn series is *strongly summable* when the union of its supports is well ordered *and* only finitely many members contribute to any one exponent. Its

sum is defined by finite addition at each exponent. For a well ordered $E \subset \Gamma_{>0}$, let E^* denote the additive monoid it generates, including 0.

Both clauses are load-bearing, and neither implies the other.

Lemma 2.2 (Support mechanism). *If $S, T \subset \Gamma$ are well ordered, then $S + T$ is well ordered and each exponent has only finitely many decompositions $s + t$. If $E \subset \Gamma_{>0}$ is well ordered, then E^* is well ordered and a fixed exponent has only finitely many representations as a finite ordered word in E — including only finitely many possible word lengths. The same conclusions hold for $S + E^*$.*

These are the Hahn–Neumann facts used throughout [22, 24]. Three consequences are used constantly. First, a complex-linear map $L : V \rightarrow W$ extends coefficientwise by $L(\sum_{\gamma} v_{\gamma} t^{\gamma}) = \sum_{\gamma} L(v_{\gamma}) t^{\gamma}$; this cannot enlarge support and requires no norm continuity. It will be applied to ordinary integration, differentiation, residue extraction, and chosen linear splittings of differential complexes. Second, for positive-support A the Neumann inversion $(1 + A)^{-1} = \sum_{q \geq 0} (-A)^q$ is strongly summable with output support in E^* . Third, $S + E^*$ is the universal support certificate: every construction below whose contributions are indexed by one source exponent in S together with a finite word in E is legitimate, and its output support lies in $S + E^*$.

Well-ordering cannot be weakened to positivity: $\{1/r : r \geq 1\}$ is a positive subset of $\mathbb{Q}$ that is not well ordered, and E^* would then fail to be well ordered.

Example 2.3 (Strong sums without valuation limits). For $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, every integer is less than ω . The strong sum $\sum_{n \geq 0} t^n = (1 - t)^{-1}$ exists, but its finite partial sums do not converge to it in the valuation topology: their nonzero tails never have valuation greater than ω . Consequently no argument of the form “the valuation improves by a fixed positive amount at each iteration” justifies any of the constructions below.

A second negative companion is equally load-bearing, and is stated here once because several proofs depend on it. “Only finitely many dependencies at a given exponent” does not mean “only finitely many exponents below it.” The latter fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$, where the interval of exponents below ω contains every positive integer. It is the former statement, supplied by lemma 2.2, that legitimises the finite-parameter replacement arguments in the residue transformation proofs of section 11 and section B. It is therefore unsafe to argue that “truncating below γ is a finite calculation”; the correct statement is that a *fixed output exponent* has finite combinatorial dependency.

2.3 The four coefficient rings, named once and never interchanged

Four coefficient rings appear below under nearly identical notation. They carry nearly identical theorem statements, and confusing them is the most likely source of a false attribution in this subject. Every theorem in this article names the ring it is stated over, and no theorem is transferred from one to another.

Ring	Notation	Defining requirement
Common-domain coherent	$\mathcal{H}(U) = \mathcal{O}(U)((t^{\Gamma}))$, $U \subset \mathbb{C}$ ordinary open	Every coefficient is holomorphic on <i>one</i> common ordinary domain U , and all share <i>one</i> well-ordered support. This is the fixed-domain (in one variable, fixed-disk) ring.

Ring	Notation	Defining requirement
Polynomial-coefficient	$\mathbb{C}[w_1, \dots, w_n]((t^\Gamma))$	Every Hahn coefficient is an ordinary polynomial; equivalently the common-domain ring with $U = \mathbb{C}^n$. This is the uniform-support target of rescaling, theorem 10.1 .
Radius-free germs	$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$	Every coefficient is an ordinary convergent germ at 0, but their convergence radii may tend to zero, so <i>no common polydisk is required</i> .
Formal	$K[[z]]$, and $\mathbb{C}[[z]]((t^\Gamma))$	No convergence of any coefficient is required. Used only for injectivity statements and for the formal residue class (12.4).

The inclusions $\mathbb{C}[w]((t^\Gamma)) \subset \mathcal{H}(\mathbb{C}^n)$ and $\mathcal{A}_n \subset \mathbb{C}[[z]]((t^\Gamma))$ are strict, and the containment $\mathcal{H}(U) \subset \mathcal{A}_n$ (for U a polydisk about 0, after taking germs) is strict as well. The running witness for strictness of the last containment is

$$H(z) = \sum_{j \geq 1} \frac{t^j}{1 - jz} \in \mathcal{A}_1, \quad (2.2)$$

whose coefficient poles at $1/j$ accumulate at 0, so that no fixed positive ordinary radius about zero carries all of its coefficient germs. Integrating it on one fixed ordinary circle as though all coefficients were holomorphic inside that circle would be invalid. It nevertheless evaluates at every $z \in \mathfrak{m}_K$. The remedy is not a uniform ordinary bound but infinitesimal rescaling; see [section 10](#).

A related warning is recorded here once. Changing the order of formal completion and Hahn extension can change the domain of the resulting expression. For instance

$$\frac{1}{z - t} = \begin{cases} \sum_{n \geq 0} t^n z^{-n-1}, & v(t) > v(z), \\ -\sum_{n \geq 0} t^{-n-1} z^n, & v(z) > v(t), \end{cases} \quad (2.3)$$

and the second expression lies in $K[[z]]$ but *not* in the unscaled radius-free ring $\mathcal{A}_1$, since its combined coefficient support in t is not well ordered. The first is an outer expansion enclosing the displaced pole; the second is a Taylor expansion on a smaller region not enclosing it. Their respective contour residues about zero are 1 and 0, and there is no contradiction: the admissible regions differ.

2.4 The perturbation residue, and why it is not a contour integral

Fix an ordinary analytic germ $f = (f_1, \dots, f_n)$ at 0 with isolated common zero, and a perturbation $F = f + E$ with $E_j \in \mathcal{A}_n$ and $v_H(E_j) > 0$ for each nonzero E_j . Zero perturbations are permitted. For $H \in \mathcal{A}_n$ the geometry manuscript defines

$$\text{Res}_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \text{res}_{(f_1^{k_1+1}, \dots, f_n^{k_n+1})} \left(H \prod_{j=1}^n E_j^{k_j} \right), \quad (2.4)$$

where res is the ordinary local Grothendieck residue, a complex-linear functional on analytic germs, extended coefficientwise, and the normalization $(2\pi i)^{-n}$ is included in res . In one variable this reads

$$\text{Res}_F(H) = \sum_{k \geq 0} (-1)^k \text{res}_{f^{k+1}}(H E^k), \quad \text{res}_{f^{k+1}}(q) = \text{Res}_{z=0} \frac{q(z) dz}{f(z)^{k+1}}. \quad (2.5)$$

The positive support of E and lemma 2.2 prove strong summability, with output support contained in $\text{supp } H + (\text{supp } E)^*$.

We state emphatically, once, what these formulas are. Equation (2.4) is a series definition, not a surcomplex contour integral. The geometry manuscript defines it algebraically precisely because no contour was available on $\mathcal{A}_n$. Supplying a contour that represents it is a separate task, and is the subject of sections 10 and 11. The sign $(-1)^{|k|}$ and the factor $(2\pi i)^{-n}$ on the contour side are the chosen normalization; they are fixed so that $\text{Res}_F(J_F) = \mu$ and $\text{Res}_F(H) = \sum_a H(a)/J_F(a)$ at simple zeros, where $J_F = \det(\partial F_i / \partial z_j)$.

Three normalizations involving $2\pi i$ are kept deliberately separate throughout, and are never silently identified.

1. **Hahn contour.** The microscopic circle functional carries the factor $1/(2\pi i)$ and the torus functional $1/(2\pi i)^n$ outside the integral, so that each equals the algebraically defined perturbation residue on the nose.
2. **Local one-variable.** The residue $\text{res}_a(Rdz)$ is the bare Laurent or partial-fraction coefficient $c_{a,1}$, after cancellations in the local finite-meromorphic germ; it is not merely the residue of one ordinary coefficient at $\text{st}(a)$. The factor $1/(2\pi i)$ again appears outside.
3. **Formal or algebraic.** In (12.4) and in the algebraic pairing (12.1) the factor $2\pi i$ is a chosen normalization matching ordinary complex integration, using the distinguished copy of $\mathbb{C}$ inside K . It is not forced by ordered-field arithmetic; over an abstract real closed field with no distinguished copy of $\mathbb{R}$ one would normalize formally or work with $I/(2\pi i)$.

3 Three topologies, and why the naive constructions fail

3.1 The full fine topology and its restriction

The full fine neighborhoods in **SC** are the balls $B(a, r) = \{z : |z - a| < r\}$ for every positive surreal r . A statement about this topology is a class-topological statement; it is not a claim that all fine-open classes form a set. The following conclusions are proved in the analysis manuscript [1, §3]; the short argument is repeated because its exact scope is crucial.

Proposition 3.1 (Full-class discreteness of sets). *Every set-sized subset of **SC** is closed and discrete in the full fine topology. Every convergent net with a set-sized index set is eventually equal to its limit. Every fine-continuous map from an ordinary connected real interval, or from an ordinary circle, into **SC** is constant. In particular, the topology induced from the full fine topology on a fixed set-sized field K is discrete.*

Proof. For a point p and a set A , the nonzero numbers $|p - a|$, $a \in A$, form a set of positive surreals. The cut $\{0 \mid |p - a| : a \in A, a \neq p\}$ supplies a positive surreal δ strictly smaller than all of them, so $B(p, \delta)$ misses $A \setminus \{p\}$. This proves closedness and discreteness. Applying the same argument to the set of nonzero distances of the values of a net from its proposed limit, eventual membership in the resulting ball forces eventual equality. Finally, the image of an ordinary connected parameter space is a set; a connected subset of a discrete space has at most one point. $\square$

This is *not* a theorem that a Hahn field with its own valuation topology is discrete. The radius supplied by the cut may lie outside the workspace. The usual real-parameter unit-circle map is not fine-continuous; calling it a contour below will therefore always mean an admissible parameterized integration object, never a fine-continuous loop.

3.2 The intrinsic topology of a fixed Hahn workspace

The internal ordered-field topology on K uses radii from R_Γ . It agrees with the natural valuation topology: monomial valuation balls refine ordered balls, and sufficiently small ordered monomial balls refine any prescribed valuation ball, since $v(x) > v(r)$ implies $|x| < r$ for $r > 0$, while $|x| < t^{\gamma+\eta}$ with $\eta > 0$ implies $v(x) > \gamma$.

Proposition 3.2 (Intrinsic total separation). *The intrinsic topology on K is nondiscrete and totally separated. The cosets $a + r\mathfrak{m}_K$, $r \neq 0$, are open and closed and separate any pair of distinct points. Every connected subset is a singleton, every continuous map from an ordinary real interval or ordinary circle to K is constant, and ordinary singular homology of K in positive degrees vanishes.*

Proof. Every neighborhood of zero contains a nonzero monomial of sufficiently large exponent, so the topology is nondiscrete. The subgroup $r\mathfrak{m}_K$ is the valuation ball $\{x : v(x) > v(r)\}$; it is open, and its complement is the union of the other open cosets, so it is clopen. For $a \neq b$ the coset $a + (b - a)\mathfrak{m}_K$ contains a and not b , so a connected subset has at most one point, and continuous images of connected ordinary spaces are singletons. Every ordinary singular simplex is therefore constant, and the singular chain complex is the direct sum over points of the singular chain complex of a point, whose positive-degree homology vanishes. $\square$

The same clopen separation works in the full fine topology, using $a + (b - a)\mathfrak{m}$ with the full class of infinitesimals. There is consequently no ordinary topological embedding $S^1(\mathbb{R}) \hookrightarrow \mathbf{SC}$ in either topology, and deleting a putative circle does not leave exactly two nontrivial ordinary connected components. The nonzero periods constructed below therefore do *not* come from that singular chain complex.

3.3 The standard-part topology

On K° — or on the affine arena $X_{a,r} = a + rK^\circ$ for $a \in K$ and $r \in R_\Gamma$ positive, via the surjection

$$q_{a,r} : X_{a,r} \longrightarrow \mathbb{C}, \quad q_{a,r}(z) = \text{st}((z - a)/r) \quad (3.1)$$

— declare the open sets to be exactly the sets $q_{a,r}^{-1}(V)$ with $V \subset \mathbb{C}$ ordinary open. This is the initial topology of the standard-part map, the *standard-part topology* of [section 1.5](#). Its fibres are indiscrete and it is not Hausdorff, but its quotient by indistinguishability is exactly the ordinary complex plane. It deliberately forgets all infinitesimal distinctions. Its role is limited but clean: it supports a coarse Jordan theorem, not microscopic separation.

Remark 3.3 (Three topologies must not be conflated). The intrinsic topology of K uses radii in R_Γ . The topology inherited by the set K from the full class $\mathbf{SC}$ also permits radii outside R_Γ and is discrete by [proposition 3.1](#). The standard-part topology is coarser still. A claim about convergence or connectedness in one of them is not automatically a claim in either of the others.

3.4 Unrestricted fundamental theorems are impossible

Proposition 3.4 (No endpoint integration for all fine-local primitives). *There is no family of linear endpoint integral functionals $I_{a,b}$, defined for all fine-locally analytic derivatives and arbitrary endpoints, satisfying $I_{a,b}(H') = H(b) - H(a)$ for every fine-locally analytic H on the full surcomplex plane.*

Proof. The class $\mathfrak{m}$ of all infinitesimals and each of its additive cosets are fine-open and fine-closed. Hence

$$q(z) = \begin{cases} 0, & z \in \mathfrak{m}, \\ 1, & z \notin \mathfrak{m} \end{cases}$$

is locally constant, so fine-locally analytic, with $q' = 0$. At $a = 0$, $b = 1$ the proposed identity gives $I_{0,1}(0) = q(1) - q(0) = 1$, contradicting linearity. $\square$

This is the obstruction proved in [1, §3]; it is *not* an obstruction to a smaller coherent calculus. A proposed Stokes theorem must restrict either the forms, the chains, or the differential. The Hahn theory of sections 6 and 7 restricts coefficient coherence; the semialgebraic theory of section 4 restricts definability; the polynomial theory of section 5 restricts to finite algebraic data.

The analysis manuscript gives a sharper diagnostic. It constructs a globally fine-locally analytic logarithm $\mathcal{L}$ on $\mathbf{SC}^\times$ with $\mathcal{L}' = 1/z$, using a selected argument cut [1, section on global logarithms]. Its restriction to ordinary points has that argument jump and is *not* a coherent primitive on $(\mathbb{C}^\times)^\sharp$. Accordingly the coherent period $\oint_{|z|=1}^H dz/z = 2\pi i$ of section 8 does not contradict the existence of $\mathcal{L}$. Applying the coherent endpoint theorem theorem 8.2 to this noncoherent global function would be an invalid change of category; a valid fundamental theorem must restrict the primitive and the integration category together.

3.5 Finite meshes cannot resolve infinitesimal tolerances

Proposition 3.5 (Finite-partition obstruction). *Let $0 = x_0 < x_1 < \cdots < x_N = 1$ be a partition in R_Γ with an ordinary finite number N of subintervals. Its mesh is at least $1/N$. For $f(x) = x$ the right and left endpoint sums differ by*

$$\sum_{j=1}^N (x_j - x_{j-1})^2 \geq \frac{1}{N}. \quad (3.2)$$

Consequently ordinary finite tagged partitions cannot make all tag choices agree to a fixed positive infinitesimal error on $[0, 1]_{R_\Gamma}$.

Proof. The increments $\Delta_j = x_j - x_{j-1}$ are positive and sum to one, so at least one is at least $1/N$. The difference of the two endpoint sums is $\sum_j \Delta_j^2$, and the algebraic identity

$$N \sum_j \Delta_j^2 - \left(\sum_j \Delta_j \right)^2 = \sum_{j < k} (\Delta_j - \Delta_k)^2 \geq 0$$

proves (3.2). For any positive infinitesimal ϵ one has $1/N > 2\epsilon$ for every ordinary N , so the two tag choices cannot both lie within ϵ of one proposed integral. $\square$

There is a related logical pitfall. A definition asserting that “all partitions of mesh less than δ have sums close to I ” becomes vacuous when δ is infinitesimal: there are no such ordinary finite partitions. One must specify an actual directed approximation system and prove it has enough elements. Note also what is *not* being claimed: a polynomial antiderivative still gives $\int_0^1 x \, dx = 1/2$, and the obstruction is to a particular unrestricted finite-mesh definition, not to all integration. Hyperfinite partitions in a nonstandard universe can be useful, but they require an additional universe, transfer structure, and a chosen relation between that universe and the surreal or Hahn workspace. *They are not supplied merely by the field operations in $\mathbf{No}$* , and taking only the ordinary standard part of such an integral would also discard the infinitesimal information sought here.

3.6 A contour avoiding every pole need not have an admissible pullback

Consider the ordinary unit-circle trace and $a = 1 + t$ with $t > 0$ infinitesimal. The function $1/(z - a)$ has no pole on that trace. Away from $z = 1$ its Hahn expansion begins with $1/(z - 1)$, but at $z = 1$ its value is $-t^{-1}$. It cannot have a single Hahn expansion with continuous coefficient functions on the entire ordinary circle: the negative-exponent coefficient would vanish at all other ordinary points but not at 1. For $a = 1 - t$ the same problem occurs.

Semialgebraic geometry distinguishes these two poles as outside and inside the full unit circle respectively. A coefficientwise contour integral based on the original macroscopic parameterization does not automatically do so. One needs a boundary-layer change of chart, a finer scale, or the algebraic pairing developed in [section 12](#). Avoidance of actual poles is necessary, but uniform admissibility of the pulled-back coefficients is an additional hypothesis. This example recurs in [section 12.3](#) as the explicit case where the two compatibility results must not be conflated.

4 Jordan separation: what survives, and in which sense

4.1 A thick-wall theorem in the standard-part topology

Theorem 4.1 (Jordan separation in the standard-part topology). *Let $J \subset \mathbb{C}$ be an ordinary Jordan curve with bounded interior D and exterior E . Fix $a \in K$ and $r \in R_\Gamma$ with $r > 0$, and use the map (3.1). Then*

$$X_{a,r} \setminus q_{a,r}^{-1}(J) = q_{a,r}^{-1}(D) \sqcup q_{a,r}^{-1}(E)$$

has exactly two connected components in the standard-part topology, and both have boundary $q_{a,r}^{-1}(J)$ relative to $X_{a,r}$. In the case $a = 0$, $r = 1$ this reads $K^\circ \setminus J^\sharp = D^\sharp \sqcup E^\sharp$ with common boundary $J^\sharp$.

Proof. For any subset $B \subset \mathbb{C}$ the subspace $q^{-1}(B)$ has precisely the saturated open sets pulled back from B , since every open set of the standard-part topology is saturated for the fibres of q and every fibre is nonempty. A separation of $q^{-1}(B)$ therefore descends to a separation of B , and conversely; so $q^{-1}(B)$ is connected exactly when B is. The ordinary Jordan theorem gives $\mathbb{C} \setminus J = D \sqcup E$ with both sets nonempty and connected, whence $q^{-1}(D)$ and $q^{-1}(E)$ are connected. They are disjoint relatively clopen subsets covering the complement, so they are exactly its components. Finally

$$\overline{q^{-1}(B)} = q^{-1}(\overline{B}), \quad \text{int}(q^{-1}(B)) = q^{-1}(\text{int } B),$$

directly from the neighborhood definition and surjectivity of q , and the ordinary boundary identities give the last assertion. $\square$

The removed object is the entire *boundary tube* $q^{-1}(J) = a + r(J + \mathfrak{m}_K)$, the monad of every point of J , and not a one-dimensional embedded curve or the thin trace $\{a + r\gamma(s) : s \in S^1(\mathbb{R})\}$ of one parameterization. A continuous coarse path whose endpoints have standard parts on different sides meets this thick wall. Several things are *not* asserted. Two infinitesimally separated points in the same boundary monad cannot be assigned different sides by this construction; different points in a fibre can lie on opposite sides of a finer-scale separator, and a finer scale is needed to distinguish two zeros in one standard-part fibre. Neither component is being called fine-connected. And “exterior” here means that the standard-part image is ordinary unbounded: all of K° is bounded by any positive infinite element of R_Γ , so we have *not* obtained an unbounded exterior in the internal ordered-field sense. This is an exact topological theorem, but it recovers only the geometry visible at the selected scale.

Example 4.2 (A boundary layer cannot be discarded silently). At $a = 0$, $r = 1$, both $1 - t$ and $1 + t$ reduce to the point 1 on the unit circle, so neither belongs to the thickened disk $\mathbb{D}^\# = q^{-1}(\mathbb{D})$, although $|1 - t| < 1 < |1 + t|$. The theorem’s interior is *not* the ordered-radius ball $\{|z| < 1\}$, and a Cauchy formula whose hypothesis is $z \in \mathbb{D}^\#$ does not automatically cover $z = 1 - t$.

This construction remains available at every infinitesimal or infinite affine scale, and it explains the topology underlying the ordinary-base coherence of [1] — without asserting that the coefficient functions of a coherent family are continuous for any of the other topologies under discussion.

4.2 The semialgebraic replacement

For this subsection let R be a real closed field containing $\mathbb{R}$ and $K = R[i]$; the case of interest is $R = R_\Gamma$. A set is *semialgebraic* if it is described by a finite Boolean combination of polynomial equations and inequalities with parameters in R . A map is semialgebraic when its graph is. A set is *semialgebraically connected* (equivalently, definably connected) when it has no separation into two nonempty relatively open semialgebraic subsets. Paths now have parameter domain the *field* interval

$$[0, 1]_R = \{s \in R : 0 \leq s \leq 1\},$$

not just the ordinary real points in it. Definable connectedness and definable path connectedness agree for semialgebraic sets; this is part of the standard finite-topology theory [4, 6].

Theorem 4.3 (Semialgebraic Jordan theorem; classical input). *Let $j : S^1(R) \rightarrow R^2$ be a semialgebraic homeomorphism onto its image J , where $S^1(R) = \{(x, y) : x^2 + y^2 = 1\}$. Then $R^2 \setminus J$ has exactly two semialgebraically connected components. They are semialgebraically path connected, one is bounded by an element of R and the other is unbounded, and J is the common boundary of both. Every semialgebraic continuous path between the two components meets J . More generally, smooth definable Jordan–Brouwer separation holds in the o-minimal manifold framework of Berarducci–Otero.*

We import this theorem, rather than claim to prove arbitrary topological Jordan separation by elementary first-order transfer. One route uses compatible finite triangulation and the combinatorial separation theorem for an embedded polygonal circle on a triangulated sphere: compactify the ambient plane semialgebraically, triangulate the pair consisting of the ambient sphere and the curve, observe that the curve is a finite subcomplex which is a circle and that the adjacent two-dimensional simplices split into two sides, and compute connectivity by chains of incident simplices. The component containing the point at infinity is the exterior; the other has bounded realization; local incidence at the curve gives the common-boundary conclusion. A path missing the curve cannot move from one relatively open definable side to the other, because its parameter interval is definably connected. The finite simplicial incidence data do not depend on which real closed field realizes them, nor on the Archimedean property of the coefficients. Alternatively, the definable Jordan–Brouwer theorem of Berarducci–Otero supplies the smooth-manifold case directly, and Woerheide’s o-minimal homology the general one; the semialgebraic triangulation framework handles the topological formulation [4, 5, 6, 30, 23]. The technical finite-topology input is being imported, not replaced by an informal appeal to transfer of arbitrary continuous maps.

Several qualifications are essential. The theorem does *not* say that those components are connected in the unrestricted intrinsic topology: by [proposition 3.2](#) all their ordinary connected components are singletons, and the nondefinable cuts that disconnect the ordered field are excluded from the definition of semialgebraic connectedness. This is why [theorem 4.3](#) does not contradict

proposition 3.1: $[0, 1]_R$ is definably connected but need not be connected as an ordinary topological space. “ R -bounded” does not mean bounded by an ordinary real number. The generalization in the last sentence refers to the differentiability and definability hypotheses of the o-minimal framework; it is not a statement about arbitrary class functions or arbitrary fine-topological subsets. And definable results concern a finite description and its real-closed extensions; they do not supply ordinary singular homology of a proper class. Finally, a semialgebraic problem involving full surreal parameters may always be placed in a real closed set-sized workspace containing those parameters, which makes this a genuine separation theorem for surcomplex data.

Example 4.4 (A microscopic circle with a genuine definable inside). For $a \in K$ and $r \in R_\Gamma$, $r > 0$, the circle $C_{a,r} = \{z \in K : |z - a| = r\}$ has semialgebraic inside $|z - a| < r$ and outside $|z - a| > r$. This remains true for $r = t^\rho$, for an infinite r , and for arbitrary a . The two points $a + r(1 - t^\eta)$ and $a + r(1 + t^\eta)$, $\eta > 0$, lie on opposite definable sides although their normalized standard parts are equal.

A distinction that will recur: the *full semialgebraic circle* $C_{a,r}$, the *coarse thick wall* $q^{-1}(J)$ of [theorem 4.1](#), and the *ordinary-parameter contour trace* $\{a + re^{i\theta} : \theta \in [0, 2\pi]_\mathbb{R}\}$ are three different objects, even when they are informally written with the same equation. The last is generally only a small subset of the first, and its parameterization is not fine-continuous.

4.3 An integer-valued index without a contour integral

For a semialgebraic loop γ avoiding b , define its index as the definable degree of

$$s \mapsto \frac{\gamma(s) - b}{|\gamma(s) - b|} \quad \text{into } S^1(R),$$

equivalently the corresponding definable homology invariant. With positive orientation it is 1 inside a semialgebraic Jordan curve and 0 outside, it takes values in $\mathbb{Z}$, and it is invariant under definable homotopies avoiding b . We write it $\text{wind}_{\text{sa}}(\gamma, b)$. *This is definable degree, not the ordinary topological degree of an ordinary real-parameter map into the intrinsic field topology.*

For piecewise polynomial or rational parameterizations there is a computationally explicit substitute. After a generic fixed rotation of the coordinate axes, write $\gamma - b = x + iy$ on each piece and form a Cauchy index; the rotation can be chosen so that no piece has identically zero y . With the convention that a jump from $-\infty$ to $+\infty$ contributes $+1$,

$$\text{wind}_{\text{sa}}(\gamma, b) = \frac{1}{2} \sum_{\text{pieces}} \text{Ind}\left(\frac{x}{y}\right), \quad (4.1)$$

with the usual half-endpoint conventions and subdivision at exceptional points; a constant piece contributes zero and may be removed first. Sturm chains compute these indices using field arithmetic and sign tests. Eisermann develops this over arbitrary real closed fields [15]; Perrucci and Roy supply corrected product formulae and a systematic treatment of boundary cases [25]. Below, every pole is disjoint from the loop, so boundary conventions are not needed, and fractional conventions for zeros on an edge or vertex are never interpreted as contour integrals through poles.

Three limitations are recorded. First, this degree is an ordinary integer, not a surreal number of turns. Second, *it does not by itself define the integral of a general differential form*: the primitive of a semialgebraic function need not be semialgebraic, and a chosen analytic expansion of a real closed field does not automatically contain all the radius-free Hahn germs of $\mathcal{A}_n$. Third, the availability of

a finite algebraic prescription does not mean arbitrary surreal inputs have computable normal forms or decidable sign tests; finite Sturm and sign algorithms use an ordered-field arithmetic oracle, and the construction becomes an algorithm only when the coefficient field and support arithmetic have effective presentations.

4.4 What Jordan separation does not supply

Separation by itself does not define an integral. Even in a definable setting, integrating a definable function can require new functions or a larger target object (see [section 14](#)). Conversely, a useful period theory can be defined on cycles with self-intersections without invoking Jordan separation at all. Below, the standard-part theorem interprets ordinary-base domains, the definable theorem interprets algebraic winding, and parameterized chains prove Stokes. *These are compatible uses, but not identifications of categories.*

5 A completely algebraic Stokes theorem for polynomial chains

There is a direct counterpart of generalized Stokes that needs neither ordinary fine paths nor infinite summation, and that lives over an arbitrary characteristic-zero coefficient field rather than over any of the four rings of [section 2.3](#). It also supplies a compatibility test for the Hahn construction of [section 6](#).

5.1 Polynomial integration on a simplex

For a characteristic-zero field L , *define* a linear functional on polynomials on the standard n -simplex by

$$\int_{\Delta^n}^{\text{pol}} u_1^{\nu_1} \cdots u_n^{\nu_n} du_1 \wedge \cdots \wedge du_n = \frac{\nu_1! \cdots \nu_n!}{(|\nu| + n)!}, \quad (5.1)$$

extended L -linearly. The geometric notation $\Delta_R^n = \{u_i \geq 0, \sum u_i \leq 1\}$ is available when L contains an ordered field R , but formula (5.1) is the definition, and the functional is therefore prior to any point set. The 0-simplex functional is evaluation. Give the simplex the ordered vertices $(0, e_1, \dots, e_n)$ and its boundary the alternating face orientation.

A *polynomial simplex* $\sigma : \Delta_R^n \rightarrow R^d$ is a polynomial map with field coefficients, and for a polynomial n -form α with coefficients in L we set $\int_{\sigma}^{\text{pol}} \alpha = \int_{\Delta^n}^{\text{pol}} \sigma^* \alpha$. Only finitely many algebraic operations are involved.

Theorem 5.1 (Polynomial-chain Stokes). *For every polynomial simplex σ and every polynomial $(n - 1)$ -form α ,*

$$\int_{\sigma}^{\text{pol}} d\alpha = \int_{\partial\sigma}^{\text{pol}} \alpha. \quad (5.2)$$

The identity extends to finite polynomial chains, and it holds over every characteristic-zero coefficient field L — in particular over every surcomplex workspace $K = \mathbb{C}((t^\Gamma))$ and over $\mathbf{SC}$ itself for finite data.

Proof. Pullback commutes with d by the polynomial chain rule, so it suffices to prove the identity on the standard simplex. Fix bounds on the degrees of the coefficients of α . Using (5.1) and the affine face maps, both sides are polynomials with rational coefficients in the finitely many coefficients

of α . Over $\mathbb{R}$, formula (5.1) is the ordinary monomial simplex integral, so ordinary Stokes proves their equality for every assignment of real coefficients. A polynomial over $\mathbb{Q}$ vanishing at every real tuple is zero, by induction on the number of variables. Hence the identity is a rational polynomial identity and holds over any characteristic-zero field. All boundary signs and face substitutions are finite, so linearity proves the chain assertion. $\square$

This proof uses classical Stokes to certify a finite algebraic identity; it does not transfer a statement quantifying over arbitrary integrable functions. Equivalent direct proofs expand the monomial factorials and cancel adjacent faces.

Example 5.2 (Infinite width, infinitesimal height). Let $L, H > 0$ be arbitrary elements of R_Γ and orient the triangle T with vertices $0, L, iH$ counterclockwise. Since $d(\bar{z} dz) = d\bar{z} \wedge dz = 2i dx \wedge dy$, polynomial Stokes gives

$$\int_{\partial T}^{\text{pol}} \bar{z} dz = iLH. \quad (5.3)$$

With $L = \omega$ and $H = \omega^{-1}$ the answer is the ordinary finite i , despite the extreme aspect ratio; the signed area functional is $LH/2$. No ordinary compactness of the entire field-valued triangle is asserted.

The construction also handles polynomial forms on triangulated polytopes whenever oriented internal faces cancel. It does *not* define an integral for every semialgebraic function, nor for a rational form whose denominator vanishes somewhere on the chain; rational contour periods will require residues or additional analytic data, which is exactly what [section 12](#) supplies.

6 A Hahn de Rham complex and coefficientwise Stokes

6.1 The differential graded algebra of globally supported Hahn forms

Let M be an ordinary smooth manifold, with corners allowed when integration is used, and let $\Omega^q(M; \mathbb{C})$ denote its ordinary smooth complex-valued differential forms. Define

$$\Omega_H^q(M) = \Omega^q(M; \mathbb{C})((t^\Gamma)). \quad (6.1)$$

An element is a family $\alpha = \sum_{\lambda \in S} \alpha_\lambda t^\lambda$ with *one global well-ordered support* S and ordinary smooth coefficient forms; smoothness up to a boundary means smooth extension in ordinary local corner charts. The word “globally supported” is load-bearing: a single well-ordered support serves the whole manifold. *This is a space of coefficient data over an ordinary manifold; it is not defined as the collection of all fine-smooth forms on a surcomplex point set. A derivative acts on α_λ , not on t^λ .* Put

$$d\alpha = \sum_{\lambda} (d\alpha_\lambda) t^\lambda, \quad \alpha \wedge \beta = \sum_{\nu} \left(\sum_{\lambda+\mu=\nu} \alpha_\lambda \wedge \beta_\mu \right) t^\nu.$$

The inner sum is finite by [lemma 2.2](#), so $\Omega_H^\bullet(M)$ is a differential graded algebra over K with $d^2 = 0$ and the graded Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^q \alpha \wedge d\beta$ for α of degree q . Ordinary smooth pullback is coefficientwise, commutes with d , and does not enlarge support. This d differentiates ordinary variables while holding every element of K fixed.

For a compact oriented ordinary q -manifold C with corners and a smooth map $\phi : C \rightarrow M$, define

$$\int_{(C, \phi)}^H \alpha = \sum_{\lambda \in S} t^\lambda \int_C \phi^* \alpha_\lambda, \quad (6.2)$$

whose support is contained in S . Extend by finite linearity to finite piecewise smooth chains. Finite subdivision, orientation reversal, ordinary reparameterization, and cancellation of oppositely oriented faces have their usual meanings. *There is no infinite interchange justified by an unstated analytic bound*: the operation is the coefficientwise extension of an ordinary linear functional, and the output lies in K even when no real-valued bound on the entire coefficient family is available. Noncompact chains could be admitted with suitable support conditions, but compact chains suffice for everything proved here.

Theorem 6.1 (Hahn-valued generalized Stokes). *Let C be a compact oriented ordinary q -manifold with corners, with its boundary orientation given by the outward-normal-first convention, let $\phi : C \rightarrow M$ be smooth, and let $\alpha \in \Omega_H^{q-1}(M)$. Then*

$$\boxed{\int_{(\partial C, \phi|_{\partial C})}^H \alpha = \int_{(C, \phi)}^H d\alpha.} \quad (6.3)$$

The pairing is K -linear in the form and additive in the chain, the same identity holds for finite piecewise smooth chains with their chain boundary, and the integral is invariant under ordinary orientation-preserving smooth reparameterization.

Proof. For each exponent λ , ordinary Stokes [17, Thm. 4.6.1], [19] gives

$$\int_{\partial C} (\phi|_{\partial C})^* \alpha_\lambda = \int_C d(\phi^* \alpha_\lambda) = \int_C \phi^*(d\alpha_\lambda).$$

Both Hahn sums have support in one well-ordered containing set, so equality of every coefficient proves (6.3). For $A = \sum_\mu a_\mu t^\mu \in K$, the coefficient of $A\alpha$ at a fixed target exponent is a finite convolution, so ordinary complex linearity of integration applies to each finite sum, proving K -linearity. For a finite chain, internal faces have equal pulled-back forms and opposite signs and cancel coefficientwise. Ordinary change of variables at each coefficient proves reparameterization invariance. $\square$

The ordinary Stokes theorem, with its chain and orientation conventions, is the classical input [19, 17]. The contribution of theorem 6.1 is the precise Hahn coefficient setting and its compatibility with strong summability. Its content is *not* an assertion that the full surcomplex plane is an ordinary manifold: the chain is ordinary, while its coefficients and its integral may be genuinely nonstandard.

6.2 A support-preserving Poincaré lemma

On an ordinary star-shaped open $U \subset \mathbb{R}^d$ containing 0, define the radial homotopy operator, for $q \geq 1$, by

$$(h\alpha)_x(v_1, \dots, v_{q-1}) = \int_0^1 s^{q-1} \alpha_{sx}(x, v_1, \dots, v_{q-1}) ds, \quad (6.4)$$

and apply it to each Hahn coefficient. Ordinary smoothness at $s = 0$ makes the result an ordinary smooth form, and the Hahn support does not increase.

Proposition 6.2 (Support-preserving Poincaré lemma). *The coefficientwise extension of (6.4) to $\Omega_H^\bullet(U)$ satisfies $dh + hd = \text{id} - c_0^*$, where c_0 is the constant map to 0. Every closed Hahn form of positive degree on U is exact, with a primitive supported in the support of the original form. A Hahn function with zero differential on a connected ordinary manifold is a single constant in K .*

Proof. Differentiate the pullback under $x \mapsto sx$ with respect to s and use the ordinary chain rule; integrating from 0 to 1 gives the displayed homotopy identity for each coefficient, equivalently the ordinary fundamental theorem applied to the radial homotopy. Every operation is complex-linear, so its coefficientwise extension preserves support. In positive degree $c_0^* = 0$. In degree zero, a zero differential makes each smooth coefficient locally constant, hence constant on a connected ordinary base. $\square$

The explicitness of the operator matters: it produces a primitive with no support enlargement whatever, which is what later support certificates require.

6.3 Cohomology and periods

Theorem 6.3 (Cohomology of the globally supported Hahn complex). *For an ordinary smooth manifold M there is a canonical K -linear isomorphism*

$$H^q(\Omega_H^\bullet(M), d) \simeq H_{\text{dR}}^q(M; \mathbb{C})((t^\Gamma)). \quad (6.5)$$

In particular a closed Hahn form is exact if and only if each of its ordinary coefficients is exact. If the ordinary Betti number b_q is finite, the right side is isomorphic to K^{b_q} and naturally to $K \otimes_{\mathbb{C}} H_{\text{dR}}^q(M; \mathbb{C})$. Integration on ordinary cycles extends the usual de Rham period pairing, and when ordinary homology in degree q is finite dimensional that pairing is nondegenerate over K .

Proof. Write $Z^q = \ker d$ and $B^q = \text{im } d$ in the ordinary complex. A Hahn form is closed exactly when every coefficient lies in Z^q ; send its class to the Hahn series of ordinary cohomology classes, discarding zero coefficients. This has support contained in the original support, and exact Hahn forms map to zero. Choose a complex-linear section of $Z^q \rightarrow Z^q/B^q$; its coefficientwise extension lifts every Hahn series of classes without enlarging support, proving surjectivity. Choose also a complex-linear right inverse of d on B^q ; if all coefficient classes vanish, applying it coefficientwise produces a Hahn primitive with support in the same containing set, so the kernel is exactly the exact Hahn forms. The auxiliary choices establish existence; the induced isomorphism is the canonical coefficient-class map. Finite convolution proves K -linearity. For finite b_q , expansion in a fixed ordinary basis identifies Hahn families with scalar extension by K , a finite union of supports being well ordered; the ordinary de Rham theorem supplies a nonsingular finite period matrix against a basis of ordinary homology, and the same matrix is nonsingular over K . $\square$

The finite-dimensional qualification matters and is not a technicality. For an infinite-dimensional V , the space $V((t^\Gamma))$ generally contains families whose coefficients span infinitely many independent directions, whereas $K \otimes_{\mathbb{C}} V$ consists of finite sums of pure tensors. *One must not silently replace the first object by the second.* Nor have we claimed that the global uniform-support assignment (6.1) is a sheaf for arbitrary infinite covers: unlike holomorphic coefficient functions on a connected domain, smooth coefficients can vanish on entire open sets, and locally chosen supports need not have well-ordered union. For example, if ψ_j are smooth bumps supported near the positive integers, then $\sum_{j \geq 1} t^{-j} \psi_j(x)$ is locally a finite Hahn family on $\mathbb{R}$ but has global exponent set $\{-j : j \geq 1\}$, which is not well ordered, so it does not belong to (6.1). All chain integrals and homotopies below have explicitly common supports or involve finitely many charts.

On an arbitrary ordinary manifold the classical de Rham theorem translates (6.5) into a period criterion: a closed Hahn form of positive degree has a global Hahn primitive exactly when all its periods over ordinary smooth cycles vanish [19].

Example 6.4 (The surviving puncture class). On $\mathbb{C}^\times$ the class of dz/z spans $H_{\text{dR}}^1(\mathbb{C}^\times; \mathbb{C})$, normalized by its ordinary unit-circle period $2\pi i$; hence $H^1(\Omega_H^\bullet(\mathbb{C}^\times)) \simeq K[dz/z]$. This is ordinary-base cohomology with enlarged coefficients. It is *not* the positive-degree singular cohomology of the totally separated intrinsic $K^\times$, which vanishes by [proposition 3.2](#): the two complexes have different bases and different chains, and there is no contradiction.

6.4 Green, divergence, and area

For a Hahn one-form $P dx + Q dy$ on an admissible oriented planar 2-chain Σ , [theorem 6.1](#) reads

$$\int_{\partial\Sigma}^H (P dx + Q dy) = \int_{\Sigma}^H (Q_x - P_y) dx \wedge dy, \quad (6.6)$$

and in complex notation, for a C^1 Hahn coefficient family F ,

$$\int_{\partial\Sigma}^H F dz = 2i \int_{\Sigma}^H \partial_{\bar{z}} F dx \wedge dy. \quad (6.7)$$

These are exact identities for the specified pullbacks. They imply Cauchy's theorem for coherent holomorphic coefficients, but they also apply to nonholomorphic coefficients. In three dimensions the same construction applied to a Hahn family of vector fields X with $d(\iota_X \text{vol}) = (\text{div } X) \text{vol}$ gives the divergence identity; defining flux by this form avoids introducing an unproved independent theory of surreal surface area and unit normals.

Example 6.5 (An infinitesimal rectangle has nonzero area). Let $A, B \in R_\Gamma$ be positive and parameterize a rectangle with side lengths A, B by the ordinary unit square. For $\alpha = -y dx + x dy$ one has $d\alpha = 2 dx \wedge dy$; the right edge contributes AB , the top edge contributes AB , and the other edges contribute zero, so

$$\int_{\partial([0,A] \times [0,B])}^H \alpha = 2AB = \int_{[0,A] \times [0,B]}^H d\alpha. \quad (6.8)$$

The chain notation denotes the indicated affine parameterization; it does not presume a fine-continuous ordinary square. For $A = t$ and $B = t^\omega$ the value is $2t^{1+\omega} \neq 0$, whereas the standard part alone would be zero, losing the geometric content of the exact identity.

7 Infinitesimally deformed chains, homotopy, and estimates

7.1 Admissible deformations and the Taylor-jet pullback

Definition 7.1 (Admissible positive-support deformation). Let M be an ordinary *compact* smooth parameter manifold with corners and let $U \subset \mathbb{R}^d$ be ordinary open. An *admissible deformation* of a smooth map $\phi_0 : M \rightarrow U$ is

$$\Phi = \phi_0 + \eta, \quad \eta = \sum_{e \in E} \eta_e t^e, \quad (7.1)$$

where $E \subset \Gamma_{>0}$ is *one* well-ordered set, so the displacement has strictly positive support, and the η_e are smooth real vector-valued coefficient functions on M , extending to a neighborhood in each corner chart. For complex coordinates use real and imaginary parts; piecewise smooth parameterizations are handled by finite subdivision.

At each ordinary parameter value η is an infinitesimal vector. *The coefficient maps need not satisfy a uniform ordinary norm bound, and no assertion of fine continuity of the real-parameter map Φ is intended*; indeed a nonconstant ϕ_0 already prevents such continuity by [proposition 3.1](#). The notation Φ describes an admissible parameterized chain, not a fine-continuous map.

For a smooth coefficient function a on U , define the *Taylor-jet pullback*

$$T_\Phi a = \sum_{\alpha \in \mathbb{N}^d} \frac{(\partial^\alpha a) \circ \phi_0}{\alpha!} \eta^\alpha, \quad d\Phi_j = d\phi_{0,j} + d\eta_j, \quad (7.2)$$

and for $\omega = \sum_{\lambda, I} a_{\lambda, I} t^\lambda dx_{i_1} \wedge \cdots \wedge dx_{i_q}$ set

$$\Phi^* \omega = \sum_{\lambda, I} t^\lambda T_\Phi a_{\lambda, I} d\Phi_{i_1} \wedge \cdots \wedge d\Phi_{i_q}. \quad (7.3)$$

The result is a form on the *ordinary parameter manifold* with Hahn coefficients. For analytic a this agrees with the infinitesimal evaluation used in the supplied manuscripts. For merely C^∞ functions, (7.2) is the *specified jet prolongation*: we do not assert ordinary convergence of its Taylor series, do not identify it with every possible fine extension of a , and do not claim a canonical prolongation of every smooth function to every fine domain. Real analyticity, when assumed below, is imposed on the *target* coefficient functions; the parameter coefficients need only be smooth.

Lemma 7.2 (Pullback and differential with a support certificate). *For $\omega \in \Omega_H^q(U)$ with support S , formula (7.3) defines an element of $\Omega_H^q(M)$ with*

$$\text{supp}(\Phi^* \omega) \subset S + E^*,$$

with only finitely many contributions at each exponent. It respects products, wedge products, restrictions to boundary faces, ordinary smooth reparameterizations of M , and the differential:

$$d(\Phi^* \omega) = \Phi^*(d\omega). \quad (7.4)$$

It also respects composition whenever the intervening ordinary coefficient maps are analytic and the same positive-support substitution hypotheses hold, and it agrees with canonical real-analytic infinitesimal evaluation of the coefficients at every ordinary parameter point.

Proof. Every term is indexed by one source exponent in S together with a finite word of error exponents in E ; differentiating η_e changes its coefficient function but not its exponent, and exterior products introduce only finitely many coordinate indices. [Lemma 2.2](#) therefore gives a well-ordered total support and only finitely many contributing words at each exponent, so each output coefficient is a finite sum of products of ordinary smooth functions, forms, and derivatives of ordinary functions, hence smooth. This proves existence and the support bound.

For functions, ordinary Taylor product identities give $T_\Phi(ab) = (T_\Phi a)(T_\Phi b)$. Differentiating (7.2) produces two groups of terms — differentiating $(\partial^\alpha a) \circ \phi_0$ contributes $d\phi_{0,j}$ and differentiating η^α contributes $d\eta_j$ — and reindexing yields

$$d(T_\Phi a) = \sum_{j=1}^d T_\Phi(\partial_j a) (d\phi_{0,j} + d\eta_j),$$

the reindexing being finite at every Hahn exponent by the first paragraph. This is the chain rule needed for (7.4), and the exterior product rule completes it. Equivalently, one may fix a target

exponent, retain the finitely many coefficient functions and Taylor degrees that can contribute, and verify the identity as an ordinary finite Taylor-jet identity; no discarded higher degree can affect that exponent. Boundary restriction and ordinary reparameterization commute with ordinary derivatives and products, hence coefficientwise. Equality at every exponent proves the Hahn identities. $\square$

Definition 7.3 (Integral on a deformed chain). For Φ as in (7.1) and an admissible form ω of degree $\dim M$, put

$$\int_{\Phi}^H \omega := \int_M^H \Phi^* \omega.$$

Boundary parameterizations are obtained by restriction to the ordinary faces of M . A finite chain is a finite linear combination of such parameterizations, whose common faces are required to agree *as parameterized Hahn data* with the appropriate orientations.

This is a definition on a parameterized object, not an attempt to integrate over its image with a fine-topological measure.

Theorem 7.4 (Stokes for Hahn-deformed chains). *Let M be compact oriented of dimension n , let Φ be an admissible deformation, and let ω be a Hahn $(n-1)$ -form with real-analytic coefficient forms on the ordinary coefficient neighborhood U . Then*

$$\boxed{\int_{\partial\Phi}^H \omega = \int_{\Phi}^H d\omega.} \tag{7.5}$$

The same identity holds for finite sums of admissible chains with matching faces, with the support bound of lemma 7.2. The integral is invariant under ordinary orientation-preserving smooth changes of parameter and under finite admissible subdivision.

Proof. By lemma 7.2 the pulled-back form is a well-defined smooth Hahn form on M and commutes with d . Apply theorem 6.1 on M :

$$\int_{\partial M}^H \Phi^* \omega = \int_M^H d(\Phi^* \omega) = \int_M^H \Phi^*(d\omega).$$

Boundary compatibility identifies the first term with the integral on $\partial\Phi$, since restriction to a face is the same as pulling back by the restricted parameterization. Ordinary change of variables at every coefficient proves reparameterization invariance, and subdivision follows by finite boundary cancellation; internal faces cancel for finite chains. $\square$

This is not merely a formal rewriting of $d^2 = 0$: it constructs a defined pairing with a chain complex and proves that the pairing intertwines d with the oriented boundary. Two cautions. First, one cannot replace “ordinary changes of parameter” by arbitrary fine-analytic reparameterizations without proving a separate substitution theorem and checking that the new pulled-back family remains admissible. Second, meromorphic forms are allowed only when their denominators have nonvanishing leading coefficient on the base image, where geometric inversion supplies holomorphic Hahn data on a common ordinary neighborhood; *passing through a leading pole is not justified by calling its displacement infinitesimal.*

7.2 Affine scales and polynomial compatibility

At an arbitrary scale write $z = a + rw$ with $a \in K$ and $r \in R_\Gamma$ positive, and suppose the pulled-back form in the w chart has admissible Hahn coefficients. Define its integral by the chart pullback; for a one-form,

$$\int_{a+r\Phi}^H F(z) dz = \int_{\Phi}^H rF(a + rw) dw. \quad (7.6)$$

For a real area form the Jacobian is r^2 , and for an arbitrary constant invertible real linear map the usual determinant appears; in several real coordinates one inserts the ordinary algebraic Jacobian and wedge factors of the affine change. Such factors lie in K and do not disturb finite convolution. *The scale factor is not optional:* a k -dimensional uniform real scaling contributes r^k to a k -form, and forgetting it would incorrectly erase microscopic areas and volumes. This is the affine contour convention of [1], here extended to deformed chains and all form degrees. Cauchy and Stokes transfer through it with their correct differential factors; a pole at $a + r\xi$ contributes the same residue after substitution, since the factor r in dz cancels the scaling of a simple-pole denominator.

A polynomial map with arbitrary Hahn coefficients need not have the specific normalization $\phi_0 + \eta$ of (7.1). It still has a finite polynomial pullback and can be integrated coefficientwise; expanding its finite polynomial coefficients and applying ordinary simplex integration gives exactly (5.1). Thus theorem 5.1 and theorem 6.1 agree on their common domain, and polynomial parameterizations are a useful enlargement beyond a single positive-support normalization.

7.3 The homotopy formula

Let $\mathcal{H} : M \times [0, 1]_{\mathbb{R}} \rightarrow U^\sharp$ be admissible in the sense of definition 7.1, with one common positive support E on the product, and write $\mathcal{H}_u$ for its restrictions. Define the degree-minus-one operator

$$\mathcal{K}\omega = \int_0^{H^1} \iota_{\partial/\partial u}(\mathcal{H}^*\omega) du, \quad (7.7)$$

the integration being coefficientwise in u and leaving a form in the M variables.

Theorem 7.5 (Hahn homotopy formula). *For every admissible form ω with support S ,*

$$\mathcal{H}_1^*\omega - \mathcal{H}_0^*\omega = d\mathcal{K}\omega + \mathcal{K}d\omega, \quad (7.8)$$

and $\mathcal{K}\omega$ has support in $S + E^$. Consequently a closed form has equal integrals on two admissibly homotopic cycles, and integrals of closed forms on closed oriented parameter manifolds are invariant under such homotopies.*

Proof. Write a coefficient of $\mathcal{H}^*\omega$ as $\beta(u) + du \wedge \gamma(u)$, both forms using only the M variables. Exterior differentiation on the product gives the ordinary coefficient identity

$$\beta(1) - \beta(0) = d_M \int_0^1 \gamma(u) du + \int_0^1 \iota_{\partial_u} d(\mathcal{H}^*\omega) du,$$

and (7.4) converts the last term. The support bound of lemma 7.2 is uniform over the ordinary compact parameter cylinder, all coefficient integrations preserve support, and finite contributions at each exponent make the calculation valid in the Hahn complex. If $d\omega = 0$, the difference is exact and its integral on a closed parameter manifold vanishes by theorem 6.1; equivalently, apply theorem 7.4 to the cylinder with the usual opposite orientations on its two ends. $\square$

Ordinary deformations of the base map are allowed if their whole image lies in the same ordinary coefficient domain; an infinitesimal deformation with fixed base is automatically of this form. A homotopy passing through a pole is *not* admissible for a form singular there, and its period can change. For nonclosed forms there is a genuine nonzero swept-region contribution: a moving boundary applied to $\bar{z} dz$ changes the integral through $d(\bar{z} dz) = 2i dx \wedge dy$. *It would be incorrect to claim that all contour integrals are infinitesimally deformation invariant.* Closedness, not mere existence of a derivative, is the controlling condition; [example 8.7](#) exhibits the effect exactly.

7.4 An order-valued parameter estimate

The coefficientwise integral does not discard the surreal order. If $A(s) = \sum_{\lambda} a_{\lambda}(s)t^{\lambda}$ has continuous real coefficients on an ordinary compact interval, one common support, and $A(s) \geq 0$ at every ordinary s , then either $A \equiv 0$ or its first nonzero coefficient function is nonnegative everywhere and positive somewhere, so its ordinary integral is positive; hence $\int^H A(s) ds \geq 0$.

Proposition 7.6 (An order-valued parameter estimate). *Write an admissible pullback as $B(s) ds$ on an ordinary interval. Suppose $m(s)$ is an ordinary nonnegative continuous function and $M \in R_{\Gamma}$ is nonnegative with $|B(s)| \leq M m(s)$ for all ordinary s . Then*

$$\left| \int^H B(s) ds \right| \leq M \int m(s) ds. \quad (7.9)$$

For an ordinary piecewise C^1 contour γ of ordinary length L and a uniformly supported continuous Hahn family F along it with $|F(\gamma(s))| \leq M$, this specializes to $|\int^H_{\gamma} F dz| \leq LM$.

Proof. If $I = \int^H B ds \neq 0$, put $u = \bar{I}/|I|$, so $|u| = 1$ and $\operatorname{Re}(uI) = |I|$. The real Hahn function $M m(s) - \operatorname{Re}(uB(s))$ is pointwise nonnegative and uniformly supported; positivity and K -linearity give (7.9). The case $I = 0$ is immediate, and piecewise regular data are treated on finitely many intervals. For the contour form take $m(s) = |\gamma'(s)|$, whose integral is L ; the integral of $M|\gamma'(s)| - \operatorname{Re}(uF(\gamma(s))\gamma'(s))$ is then $LM - |I|$. $\square$

This is the estimate proved in the analysis manuscript, here given as a numbered order-valued statement, and it shows why a coefficientwise construction retains an exact surreal — rather than merely standard-part — inequality. The formulation deliberately avoids assuming that the speed $|\Phi'(s)|$ possesses a globally uniform smooth Hahn expansion through every possible zero of its leading speed. Ordered estimates do not imply convergence of set-indexed nets in the full fine topology.

8 Cauchy theory as a period theory

Throughout this section the coefficient ring is the *common-domain coherent ring* $\mathcal{H}(U) = \mathcal{O}(U)((t^{\Gamma}))$ of [section 2.3](#), for an ordinary open $U \subset \mathbb{C}$. Every theorem below is stated over that ring and over no other; the radius-free ring $\mathcal{A}_n$ is treated separately in [sections 10](#) and [11](#), and none of the statements here is transferred to it.

8.1 Coherent holomorphy and the vanishing theorem

An element $F = \sum_{\lambda \in S} f_\lambda t^\lambda \in \mathcal{H}(U)$ has one well-ordered support S and ordinary holomorphic coefficients $f_\lambda \in \mathcal{O}(U)$ on *one* common U . At $z = c + \varepsilon \in U^\sharp$, evaluation means

$$F^\sharp(c + \varepsilon) = \sum_{\lambda, n \geq 0} t^\lambda \frac{f_\lambda^{(n)}(c)}{n!} \varepsilon^n, \quad (8.1)$$

whose support is contained in $S + (\text{supp } \varepsilon)^*$, so it is strongly defined. Differentiation acts on the f_λ . The injectivity, composition, and gluing properties of $^\sharp$ are supplied in [1, §5]. For an ordinary piecewise C^1 path γ in U ,

$$\int_\gamma^H F(z) dz = \sum_\lambda t^\lambda \int_\gamma f_\lambda(z) dz, \quad (8.2)$$

and the support cannot increase. In real coordinates

$$d(F dz) = 2i \partial_{\bar{z}} F dx \wedge dy = 0, \quad (8.3)$$

the last equality being coefficientwise ordinary holomorphy: the Cauchy–Riemann equation holds for each coefficient, not for an arbitrary fine-local function.

Theorem 8.1 (Coherent Cauchy theorem). *Let $F \in \mathcal{H}(U)$. If γ_0 is an ordinary piecewise smooth closed cycle null-homologous in U , then $\oint_{\gamma_0}^H F dz = 0$. If an admissible closed contour bounds a finite admissible 2-chain on which the coefficient functions of F are holomorphic, its period is likewise zero. In particular, for a Hahn-deformed closed contour $\gamma = \gamma_0 + \eta$ in the sense of [definition 7.1](#), the period equals the period of γ_0 , whether or not that period is zero; and every closed contour has zero period when U is simply connected.*

Proof. For an ordinary contour apply ordinary Cauchy to every coefficient, or use a smooth filling chain with (8.3) in [theorem 6.1](#). The bounding-chain assertion is [theorem 7.4](#) applied to the closed form $F dz$. For a deformed contour the homotopy $\gamma_u = \gamma_0 + u\eta$, $0 \leq u \leq 1$, is admissible with the same positive error support, so [theorem 7.5](#) applies to the closed form $F dz$; alternatively [theorem 8.2](#) identifies the two periods directly. The simply connected case follows from ordinary complex homology or from coefficientwise primitives. $\square$

Null-homology is a genuine hypothesis. On $U = \mathbb{C}^\times$ the coherent form dz/z has unit-circle period $2\pi i$, so “closed contour” alone does not imply a vanishing integral. Conversely, self-intersections do not obstruct Cauchy’s theorem when the cycle is null-homologous: neither the contour nor a filling chain need be embedded, and the Jordan theorem is not needed for this vanishing statement. The Jordan theorem is useful when identifying a particular simple boundary with its interior, or when turning winding into an inside/outside count.

8.2 Exact transport of an open contour

For $c \in U$ and $\varepsilon \in \mathfrak{m}_K$ define the infinitesimal primitive increment

$$T_F(c, \varepsilon) = \sum_{n \geq 0} \frac{F^{(n)}(c)}{(n+1)!} \varepsilon^{n+1}, \quad (8.4)$$

which involves no choice of global primitive.

Theorem 8.2 (Exact endpoint-transport formula). *Let $\gamma_0 : [a, b]_{\mathbb{R}} \rightarrow U$ be ordinary piecewise smooth, let $F \in \mathcal{H}(U)$, and let $\Gamma = \gamma_0 + \eta$ be an admissible complex deformation, piecewise on the same finite partition. Then*

$$\boxed{\int_{\Gamma}^H F dz - \int_{\gamma_0}^H F dz = T_F(\gamma_0(b), \eta(b)) - T_F(\gamma_0(a), \eta(a))}. \quad (8.5)$$

No simple-connectedness assumption on U is required. *In particular a closed deformed contour with matching endpoint displacement has exactly the period of its ordinary base contour, not merely the same standard part.*

Proof. Differentiate $T(s) = T_F(\gamma_0(s), \eta(s))$ coefficientwise. The derivative of the $F^{(n)}$ factor gives

$$\gamma'_0(s) \sum_{n \geq 0} \frac{F^{(n+1)}(\gamma_0(s))}{(n+1)!} \eta(s)^{n+1} = \gamma'_0(s) (F^{\sharp}(\Gamma(s)) - F(\gamma_0(s))),$$

and the derivative of the displacement gives $\eta'(s) F^{\sharp}(\Gamma(s))$, so

$$T'(s) = F^{\sharp}(\Gamma(s)) \Gamma'(s) - F(\gamma_0(s)) \gamma'_0(s).$$

All series have support in $\text{supp } F + E^*$, and [lemma 2.2](#) justifies every reindexing. Integrating the ordinary parameter coefficientwise and applying the ordinary fundamental theorem to each coefficient gives (8.5). Splitting into finitely many smooth pieces introduces endpoint terms which cancel at the joins; for a closed contour the two remaining terms coincide.

Equivalently, set $\gamma_u(s) = \gamma_0(s) + u\eta(s)$ and use the identity

$$\partial_u(F(\gamma_u) \partial_s \gamma_u) = \partial_s(F(\gamma_u) \partial_u \gamma_u), \quad (8.6)$$

all products and derivatives being admissible by [lemma 7.2](#); integrating coefficientwise over the ordinary rectangle $[0, 1] \times [a, b]$ and Taylor expanding identifies the two endpoint terms as (8.4). These derivatives differentiate the ordinary coefficient functions of the parameter, not a putative fine-continuous path. $\square$

Consequently the analysis manuscript's endpoint integral is the value of *every* admissible infinitesimal deformation of its base contour with those endpoints; its endpoint correction is not an arbitrary prescription. A primitive is not required, because only the homotopy strip and not an arbitrary filling disk is used. In contrast, choosing a different ordinary homology class can still change the answer by a period: a coherent global primitive eliminates those periods, merely fine-local primitives do not. The corresponding fundamental theorem for finite nonstandard endpoints $z_j = a_j + \varepsilon_j$ reads

$$-T_F(a_1, \varepsilon_1) + \int_{\gamma}^H F dz + T_F(a_2, \varepsilon_2) = \text{difference of a coherent primitive at the endpoints}, \quad (8.7)$$

which does not contradict [proposition 3.4](#) because the class of admissible primitives has been restricted.

Example 8.3 (The logarithm obstruction has not disappeared). The global fine-locally analytic logarithm $\mathcal{L}$ on $\mathbf{SC}^{\times}$ with $\mathcal{L}' = 1/z$ uses a selected argument cut and is *not* a coherent primitive on $(\mathbb{C}^{\times})^{\sharp}$. Accordingly $\oint_{|z|=1}^H dz/z = 2\pi i$ does not contradict its existence. Applying [theorem 8.2](#) to this noncoherent global function would be an invalid change of category.

8.3 Winding at a displaced center, and the Cauchy formula

Let $a = a_0 + \xi$ with $a_0 \in \mathbb{C}$, $\xi \in \mathfrak{m}_K$, and a_0 not on the ordinary contour γ_0 . On a common ordinary collar of the contour,

$$\frac{1}{z - a} = \sum_{n \geq 0} \frac{\xi^n}{(z - a_0)^{n+1}}$$

is an admissible Hahn family. Higher-pole terms are ordinary derivatives and have zero periods.

Theorem 8.4 (Winding stability at one scale). *Let γ_0 be an ordinary closed piecewise smooth contour, let $c \notin \gamma_0$, let η be an admissible closed positive-support deformation, and let $\xi \in \mathfrak{m}_K$, $a \in K$, $r \in R_\Gamma$ with $r > 0$. Set $\Gamma = a + r(\gamma_0 + \eta)$ and $b = a + r(c + \xi)$. Then the integrand is coherent on an ordinary neighborhood of the base contour and*

$$\frac{1}{2\pi i} \oint_{\Gamma}^H \frac{dz}{z - b} = \text{wind}(\gamma_0, c) \in \mathbb{Z}. \quad (8.8)$$

Proof. The affine factor r cancels by (7.6), reducing to $a = 0$, $r = 1$. Two proofs are available. By theorem 8.2 the closed period is unchanged under deformation from γ_0 to $\gamma_0 + \eta$, and on the ordinary base the $n = 0$ term of the expansion has integral $2\pi i \text{wind}(\gamma_0, c)$ while each $n > 0$ term is an ordinary derivative and has zero integral; strong summability justifies coefficientwise integration. Alternatively, on the parameter cylinder set $g_u(s) = \gamma_0(s) - c + u(\eta(s) - \xi)$. Its leading coefficient never vanishes, compactness of the ordinary contour supplying a positive ordinary separation from c , so $1/g_u$ is a uniformly supported Hahn family; direct differentiation gives $\partial_u(\partial_s g_u/g_u) = \partial_s(\partial_u g_u/g_u)$, whose integral along the closed parameter is zero coefficient by coefficient, so the period at $u = 1$ equals the ordinary winding integral at $u = 0$. $\square$

Thus an admissible contour has an exact *integer* period index, not an unspecified surreal winding number. The theorem treats only points whose standard part avoids the base curve; a point reducing to the curve lies in its boundary tube and requires a finer chart, an explicit detour, or the separate algebraic definition of section 12. This is precisely the hypothesis that fails for $a = 1 \pm t$ in section 3.6.

Theorem 8.5 (Cauchy formula at displaced points and on deformed contours). *Let D be a bounded ordinary Jordan domain with piecewise smooth positively oriented boundary and $\overline{D} \subset U$, let $F \in \mathcal{H}(U)$, and let Γ be either ∂D or an admissible closed infinitesimal deformation of it. For $z = c + \varepsilon$ with $c \in D$, $\varepsilon \in \mathfrak{m}_K$, and every $n \geq 0$,*

$$(F^{(n)})^\sharp(z) = \frac{n!}{2\pi i} \oint_{\Gamma}^H \frac{F(\zeta)}{(\zeta - z)^{n+1}} d\zeta. \quad (8.9)$$

The corresponding statement holds after any admissible affine rescaling.

Proof. Expand

$$(\zeta - c - \varepsilon)^{-n-1} = \sum_{k \geq 0} \binom{n+k}{k} \varepsilon^k (\zeta - c)^{-n-k-1}.$$

The ordinary point c has positive ordinary distance from the compact boundary, so all coefficient functions are holomorphic on one common collar, and lemma 2.2 permits multiplication by F and coefficientwise integration. Ordinary Cauchy gives the coefficient $f_\lambda^{(n+k)}(c)/k!$, yielding $\sum_{\lambda, k} t^\lambda f_\lambda^{(n+k)}(c) \varepsilon^k/k!$, which is (8.1) for $F^{(n)}$. The support is contained in $\text{supp } F + (\text{supp } \varepsilon)^*$. The kernel form is closed on the collar, so the deformed case follows from theorem 8.2; affine substitution gives the rescaled statement with the ordinary derivative and differential factors. $\square$

This proves a formula for genuinely displaced points and contours. It does *not* claim that the deformed trace is the fine-topological boundary of a fine-connected interior, and by [example 4.2](#) its hypothesis $c \in D$ is not the same as $|z| < 1$ for the unit disk.

On a simply connected ordinary domain, normalized ordinary primitives of all f_λ assemble into a coherent primitive with support in S ; on a general connected domain such a primitive exists exactly when all ordinary closed-cycle Hahn periods vanish, and all coherent primitives of one function differ by a single constant in K [[1](#), §6]. Morera’s criterion holds for common-support continuous coefficient families, since vanishing Hahn integrals around every ordinary triangle means vanishing of every ordinary coefficient integral.

8.4 Moving poles and actual zero clusters

Let $F, G \in \mathcal{H}(U)$ with $G \neq 0$, and suppose the ordinary leading coefficient function g_0 of G after monomial normalization has no zero on the ordinary boundary of a bounded Jordan domain D with $\overline{D} \subset U$ and finitely many zeros inside. The preparation and moving-pole theorem of [[1](#)] gives

$$\frac{1}{2\pi i} \oint_{\partial D}^H \frac{F}{G} dz = \sum_{b \in D^\#} \text{Res}_b \left(\frac{F}{G} dz \right), \quad (8.10)$$

the sum being over *actual surcomplex poles*, not merely their standard parts. The sum is finite; its summands may be infinite. As fixed in [section 2.4](#), the residue at b is the coefficient of $(z - b)^{-1}$ in the local finite-meromorphic germ *after cancellations*, and is not merely the residue of one ordinary coefficient at $\text{st}(b)$.

The decisive local reduction is worth recording, because it is the mechanism reused in [theorem 10.5](#). Preparation near a leading zero writes G as a coherent unit times a monic polynomial P whose lower coefficients are infinitesimal; division writes $F/G = J + R/P$ with J coherent holomorphic and $\deg R < \deg P$; the polynomial has finitely many infinitesimal roots, and rational partial fractions give $R/P = \sum_b \sum_{j=1}^{m_b} A_{b,j} (z - b)^{-j}$, where m_b is the order of b as a root of P . On a small ordinary protecting circle only the $j = 1$ terms have nonzero coefficientwise period, contributing $2\pi i A_{b,1}$, and the holomorphic part contributes zero. This argument is valid in higher rank because the expansions are strong sums, not sequential limits. It is stronger than calling a coefficientwise residue at the reduced center “the residue” and leaving its relation to the split points unspecified.

Corollary 8.6 (Residues on a deformed protecting contour). *Formula (8.10) remains true when ∂D is replaced by an admissible closed infinitesimal deformation lying in a coefficient collar on which g_0 is nonzero. The right side is still the divisor selected by the ordinary interior $D^\#$.*

Proof. The fraction is Hahn-holomorphic on that collar, so its closed period is unchanged by [theorem 8.2](#). $\square$

This describes the selected divisor without inventing a fine-connected interior of the deformed trace. The same theorem applied to F'/F gives the argument principle: its normalized period is the total zero multiplicity, or zeros minus poles for a fraction, with ordinary integer multiplicities. Positive-support perturbations preserving the leading function preserve the total multiplicity in each ordinary monad [[1](#), [2](#)]; individual roots can nevertheless split at finer scales, which is the subject of [sections 9](#) and [10](#).

Example 8.7 (A deformed ellipse and its exact area). Let $\varepsilon, \delta \in \mathfrak{m}_K \cap R_\Gamma$ and map the ordinary unit disk by $\Phi(x, y) = (1 + \varepsilon)x + i(1 + \delta)y$, with boundary $\Gamma(\theta) = (1 + \varepsilon)\cos\theta + i(1 + \delta)\sin\theta$. Since the pullback of $dx \wedge dy$ is $(1 + \varepsilon)(1 + \delta)dx \wedge dy$, [theorem 7.4](#) applied to $\bar{z}dz$ gives

$$\oint_{\Gamma}^H \bar{z}dz = 2\pi i(1 + \varepsilon)(1 + \delta) = 2\pi i(1 + \varepsilon + \delta + \varepsilon\delta). \quad (8.11)$$

Thus an infinitesimal deformation changes a nonclosed form's integral by a genuine infinitesimal, *including a second-order mixed term*. Setting $\delta = \varepsilon$ gives $2\pi i(1 + \varepsilon)^2$; further affine scaling by r multiplies by r^2 ; translation adds an exact one-form with zero closed period. In contrast $z^q dz$ is exact for every integer $q \geq 0$ and integrates to zero over both the circle and the ellipse. Writing $R = 1 + t$ and $S = 1 + t^\omega$ gives $2\pi i(1 + t + t^\omega + t^{1+\omega})$, a value visible only if all valuation ranks are retained. For the undeformed circle of radius $r \in R_{\Gamma, > 0}$ the same identity reads $\oint_{|z|=r}^H \bar{z}dz = 2\pi i r^2 = 2i \int_{r\mathbb{D}}^H dx dy$, giving a nonzero infinitesimal area for $r = t^\rho$ and an infinite one for infinite r . No measure on all fine-topological subsets is asserted.

Example 8.8 (Separating residues invisible at the coarse scale). Let $t = \omega^{-1}$ and $R(z) = 1/(z^2 - t^2)$. Its residues at $\pm t$ are $\pm 1/(2t)$, so

$$\frac{1}{2\pi i} \oint_{|z|=1}^H R(z) dz = 0 :$$

both poles reduce to zero and the coarse contour sees only their cancelling sum. Choose $r = t^\rho$ with $\rho > 1$ — *including a higher Archimedean class such as $\rho = \omega$* — and use the contour $z = t + rw$, $|w| = 1$ in the ordinary parameter. Then

$$R(z) dz = \frac{dw}{w(2t + rw)} = \frac{1}{2t} \sum_{k \geq 0} \left(-\frac{r}{2t}\right)^k w^{k-1} dw, \quad (8.12)$$

whose relative support is positive because $v(r/t) = \rho - 1 > 0$. Its normalized period is $1/(2t)$, so the actual integral is $\pi i/t$; the analogous circle about $-t$ gives $-\pi i/t$. The same support-controlled calculus therefore resolves the two infinite residues individually and recovers their cancellation on the larger contour. For $z/(z^2 - t^2)$ the two residues are $1/2$ and the coarse normalized period is 1. No all-scale coherence hypothesis on any transcendental function was used: a contour only needs a valid representation at its chosen scale, so the all-scale rigidity theorems of [\[1\]](#) pose no obstruction to this local multiscale calculus.

9 Affine scales, microscopic contours, and compatible charts

9.1 A precise compatibility criterion

Proposition 9.1 (Agreement on admissible overlaps). *Suppose two chart descriptions of the same parameterized oriented chain produce pointwise equal pulled-back differential forms at every ordinary parameter point, and both pulled-back forms are common-support Hahn families with smooth coefficients. Then their integrals agree. The same holds for a finite collection of chart pieces with matching oriented faces.*

Proof. Enlarge the workspace to contain both supports. Uniqueness of Hahn normal forms makes each coefficient of the difference vanish at every parameter point, so the coefficient forms are equal; their ordinary integrals agree, and so do their Hahn sums. For finitely many pieces use additivity and cancellation of matching faces. $\square$

This criterion is intentionally stronger than equality of the set-theoretic images of two curves: a chain includes multiplicity, orientation, and the pulled-back differential. It is also weaker than requiring one global chart for every scale, as finitely many verified overlap identities suffice. What is *not* justified is a patchwork of arbitrary local fine germs for which common-support pullbacks and their identities have not been established; equality of bare fine germs in unrelated, unverified summation descriptions is not a substitute.

9.2 Macroscopic and microscopic winding resolve different information

A macroscopic contour whose reduction avoids a cluster assigns the same winding to every point of that cluster. A contour centered on one actual pole and rescaled below the finite separation distances can resolve an individual contribution. This requires no surreal-valued winding number: ordinary integer winding numbers in different admissible charts suffice.

Example 9.2 (One coarse contour, two microscopic contours). The poles of $e^z/(z^2 - t)$ are $\pm t^{1/2}$, with residues $e^{t^{1/2}}/(2t^{1/2})$ and $-e^{-t^{1/2}}/(2t^{1/2})$. These are individually infinite, but their sum is

$$\frac{\sinh(t^{1/2})}{t^{1/2}} = \sum_{n \geq 0} \frac{t^n}{(2n+1)!}. \quad (9.1)$$

The ordinary unit circle and the protecting microscopic circle $|z| = t^{1/4}$ both have integral $2\pi i$ times (9.1). A circle centered at $t^{1/2}$ of radius t isolates the positive pole: in the coordinate $z = t^{1/2} + tw$ the second pole sits at $w = -2t^{-1/2}$, an infinite normalized distance. Each integral is computed in its own admissible chart, and the outer integral is the sum of the two inner residues even though the individual residues are infinite. *There is no assertion that these ordinary-parameter traces constitute ordinary topological circles in the fine plane.*

Example 9.3 (Two poles in different valuation ranks). Let $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ with its inherited surreal order and put $R(z) = 1/(z - t) + 1/(z - t^\omega)$, both residues being 1. Three positively oriented microscopic circles give

$$\frac{1}{2\pi i} \oint_{|z|=t^\rho}^H R(z) dz = \begin{cases} 2, & \rho = \frac{1}{2}, \\ 1, & \rho = 2, \\ 0, & \rho = 2\omega. \end{cases} \quad (9.2)$$

For the middle circle, t^ω lies inside but t lies outside, because $\omega > 2 > 1$ reverses the magnitude order of the monomials; the last circle is smaller than both pole scales. Each assertion follows by geometric expansion in the appropriate ratio, or by algebraic winding and [theorem 12.2](#). *A single real-valued valuation that discards the higher rank cannot retain all of these distinctions.*

10 Radius-free germs become coherent after microscopic rescaling

From here to the end of [section 11](#) the coefficient ring is the *radius-free germ ring*

$$\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$$

of [section 2.3](#), and the target of rescaling is the polynomial-coefficient ring $\mathbb{C}[w_1, \dots, w_n]((t^\Gamma))$. No theorem of [section 8](#), which is stated over $\mathcal{H}(U)$, is applied to $\mathcal{A}_n$ before the rescaling theorem has moved the data into a common-domain ring.

10.1 The missing common-radius problem

An element $H = \sum_{\lambda} h_{\lambda}(z)t^{\lambda} \in \mathcal{A}_n$ has ordinary convergent germs as coefficients, with no requirement that all h_{λ} extend to the same ordinary polydisk. The witness (2.2) is legitimate, but its coefficient poles at $1/j$ prevent a common ordinary disk of positive radius, so an ordinary fixed-circle coefficient integral at zero is unavailable; applying theorem 8.5 to it on a fixed ordinary circle would leave a gap. Evaluation at infinitesimal points is nevertheless strongly defined. The remedy is rescaling.

Theorem 10.1 (Positive rescaling gives polynomial coefficient families). *Let $\rho_1, \dots, \rho_n \in \Gamma_{>0}$, put $r_j = t^{\rho_j}$, and write $rw = (r_1w_1, \dots, r_nw_n)$. The Taylor substitution*

$$\sigma_r(H)(w) := H(rw) = \sum_{\lambda, \alpha} h_{\lambda, \alpha} t^{\lambda + \alpha \cdot \rho} w^{\alpha} \quad (10.1)$$

defines an injective K -algebra homomorphism

$$\sigma_r : \mathcal{A}_n \longrightarrow \mathbb{C}[w_1, \dots, w_n]((t^{\Gamma})).$$

Every fixed Hahn coefficient of (10.1) is an ordinary polynomial; the support is contained in $\text{supp } H + \{\rho_1, \dots, \rho_n\}^$; the image has coherent evaluation on the thickening of every ordinary w -domain; and $\partial_{w_j} \sigma_r(H) = r_j \sigma_r(\partial_{z_j} H)$. More generally, for $a \in \mathfrak{m}_K^n$ the translated substitution*

$$H(a_1 + r_1w_1, \dots, a_n + r_nw_n) \in \mathbb{C}[w_1, \dots, w_n]((t^{\Gamma})) \quad (10.2)$$

has support contained in $\text{supp } H + (E_a \cup \{\rho_1, \dots, \rho_n\})^$, where $E_a = \bigcup_i \text{supp}(a_i)$, and it evaluates correctly at every finite tuple $w \in (K^{\circ})^n$.*

Proof. Write the Taylor germ $h_{\lambda}(z) = \sum_{\alpha} h_{\lambda, \alpha} z^{\alpha}$. At a fixed Hahn exponent ν , the possible contributions satisfy $\lambda + \alpha_1 \rho_1 + \dots + \alpha_n \rho_n = \nu$. Lemma 2.2 gives finitely many source exponents λ and finitely many finite words in the positive alphabet $\{\rho_1, \dots, \rho_n\}$, hence finitely many multi-indices α contributing at ν ; the coefficient at ν is therefore an ordinary polynomial, and the total support is well ordered. Taylor multiplication and differentiation are valid at each exponent because the relevant regroupings are finite, which gives the algebra and derivative statements. The Taylor embedding $\mathcal{A}_n \hookrightarrow K[[z]]$ is injective, and diagonal substitution $z_j = r_j w_j$ is injective in that formal ring because $r_j \neq 0$; hence σ_r is injective. Polynomial coefficients are holomorphic on every ordinary domain, and evaluation at a finite normalized point agrees with evaluation of H at the infinitesimal point rw by the same finite-contribution Taylor identities. For (10.2), first translate by a using the strongly summable Taylor translation of [2, §3], whose support lies in $\text{supp } H + E_a^*$, then repeat the argument; evaluation at finite $w = c + \varepsilon$ is justified by adjoining the positive supports of the infinitesimal parts. $\square$

No coefficient radii enter this proof. The argument in fact works for arbitrary *formal* complex coefficient series at infinitesimal arguments, although the ring of interest here is the analytic-germ ring. Its analytically relevant conclusion is not ordinary convergence of the original germs on a uniform disk, but a *new coherent representation in an infinitesimal chart*; it neither establishes nor requires all-scale coherence of the original function. For (2.2) the substitution $z = t^{1/4}w$ gives $H(t^{1/4}w) = \sum_{j \geq 1} \sum_{q \geq 0} j^q t^{j+q/4} w^q$, at a fixed exponent a finite sum, so the singularities of the original ordinary coefficient representatives no longer obstruct coefficientwise contour integration in the w -plane. The resulting coefficients need not have any uniform degree or growth bound as the Hahn exponent varies.

Convention. For $r = t^\rho$ the symbol $\{|z| = r\}$ in a Hahn integral always means the ordinary parameterized circle $z = re^{i\theta}$, $0 \leq \theta \leq 2\pi$, with its pulled-back differential. *It does not mean the full class of points of that modulus equipped with a fine-topological integration theory*, and it is not the full semialgebraic circle of [example 4.4](#).

10.2 Microscopic Cauchy formulae over $\mathcal{A}_n$

Theorem 10.2 (Cauchy formula for a radius-free germ). *Let $H \in \mathcal{A}_1$ and $a \in \mathfrak{m}_K$. Choose $\rho > 0$ with $\rho < v(a)$ when $a \neq 0$, and put $r = t^\rho$; if $a = 0$, any $\rho \in \Gamma_{>0}$ is allowed. Then for every $k \geq 0$*

$$H^{(k)}(a) = \frac{k!}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{(z-a)^{k+1}} dz. \quad (10.3)$$

Proof. By [theorem 10.1](#), $G(w) = H(rw)$ lies in $\mathbb{C}[w]((t^\Gamma)) \subset \mathcal{H}(\mathbb{C})$ with ordinary entire polynomial coefficients, and the normalized point a/r is infinitesimal. Apply [theorem 8.5](#) to G on the ordinary unit circle. Since $G^{(k)}(a/r) = r^k H^{(k)}(a)$ and $dz = r dw$, the scaling factors yield (10.3). $\square$

The same proof gives a product-torus Cauchy formula for $H \in \mathcal{A}_n$, using one sufficiently large infinitesimal circle around each coordinate of the evaluation point. *A product of n circles is a real n -dimensional cycle, the distinguished boundary of a polydisk; it is not the real $(2n-1)$ -dimensional boundary of that polydisk.* There is also a local primitive theorem: ordinary germwise integration applied coefficientwise makes $D : \mathcal{A}_1 \rightarrow \mathcal{A}_1$ surjective with kernel K , and the ordinary analytic Poincaré homotopy applied coefficientwise makes the analytic differential-form complex of $\mathcal{A}_n$ exact in positive degrees. Thus microscopic periods of nonsingular forms vanish while poles supply residue obstructions. *These assertions use the prescribed Hahn-germ algebra, not all fine-analytic functions.*

10.3 Protecting radii and one-variable residues

Lemma 10.3 (A radius protecting an entire one-variable cluster). *Let $F = f + E \in \mathcal{A}_1$, where f has ordinary order $m \geq 1$ at zero, say $f(z) = z^m u(z)$ with $u(0) \neq 0$, and $\delta = v_H(E) > 0$ (with $\delta = +\infty$ if $E = 0$). Choose $\rho > 0$ with $m\rho < \delta$ and put $r = t^\rho$; if $E = 0$ every positive ρ works. Then $r^{-m}F(rw)$ has leading coefficient $u(0)w^m$ and no zeros on the ordinary unit-circle parameter, and every infinitesimal zero a of F satisfies $v(a) > \rho$.*

Proof. The higher powers in $u(rw) - u(0)$ have positive valuation, and every term of $r^{-m}E(rw)$ has valuation at least $\delta - m\rho > 0$; this proves the leading-coefficient and boundary assertions. For a nonzero infinitesimal zero with $v(a) \leq \rho$, the term $f(a)$ has valuation $m v(a) \leq m\rho < \delta$ while $E(a)$ has valuation at least δ , so they cannot cancel. The zero $a = 0$ satisfies the conclusion by convention. Divisibility of Γ supplies a strictly smaller positive ρ in the nonzero case, even for arbitrary rank. $\square$

The interpretation is a *protecting-radius condition*: the chosen circle must be infinitesimally large relative to the perturbation scales after accounting for the order m . Every zero in the originating infinitesimal cluster lies strictly inside the protecting circle.

Theorem 10.4 (Microscopic circle realization of the one-variable perturbation residue). *Let $F = f + E \in \mathcal{A}_1$ and $H \in \mathcal{A}_1$ be as in [lemma 10.3](#), with $r = t^\rho$ and $m\rho < v_H(E)$. Then the*

microscopic circle functional is well defined and

$$\boxed{\frac{1}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{F(z)} dz = \text{Res}_F(H),} \quad (10.4)$$

with Res_F the one-variable perturbation residue (2.5). The integral means substitution $z = rw$ followed by coefficientwise integration over the ordinary unit circle in w , and it is independent of every choice of ρ satisfying the displayed bound. Its support is contained in $\text{supp } H + (\text{supp } E)^*$.

Proof. Write $f(z) = cz^m + O(z^{m+1})$, $c \neq 0$. By theorem 10.1 both $H(rw)$ and $F(rw)$ have polynomial Hahn coefficients, and $r^{-m}F(rw) = cw^m + P(w)$ with $v_H(P) > 0$: higher terms of f acquire positive relative exponents $(j - m)\rho > 0$, and the perturbation terms have relative exponent at least $v_H(E) - m\rho > 0$. On an ordinary annulus around $|w| = 1$ the leading coefficient cw^m is nowhere zero, so geometric inversion gives a common-support holomorphic Hahn family — an element of $\mathcal{H}$ of that annulus — for $rH(rw)/F(rw)$. Expanding instead around $f(rw)$,

$$\frac{rH(rw)}{F(rw)} = \sum_{k \geq 0} (-1)^k \frac{rH(rw)E(rw)^k}{f(rw)^{k+1}},$$

which is strongly summable on the annulus since $E(rw)/f(rw)$ has strictly positive Hahn valuation there. Coefficientwise integration extracts the coefficient of w^{-1} . For an ordinary germ q the formal Laurent change of variable gives

$$[w^{-1}] \frac{rq(rw)}{f(rw)^{k+1}} = [z^{-1}] \frac{q(z)}{f(z)^{k+1}},$$

an identity that holds for a Hahn family of germs as well, by finite contributions after rescaling: in a term $g(z)dz$ of the ordinary meromorphic germ expansion, scaling sends $z^j dz$ to $r^{j+1}w^j dw$, and the circle residue retains only $j = -1$, with no surviving scale factor. Applying this to $q = HE^k$ and summing the strongly summable family gives exactly (2.5). Since the right side does not involve ρ , radius independence follows, and the support certificate is the one carried by (2.5). $\square$

Radius independence here is an *exact residue identity*. It does not require a convergent sequence of radii or a fine-continuous homotopy between scales whose ratio might be infinite.

Theorem 10.5 (Radius-free moving-pole residue theorem). *Let $N, G \in \mathcal{A}_1$ with $G \neq 0$. Normalize its leading exponent and choose a protecting microscopic circle as in lemma 10.3 when its leading germ vanishes at zero; if the leading germ is a unit, choose any sufficiently small infinitesimal circle. Then*

$$\frac{1}{2\pi i} \oint_{|z|=r}^H \frac{N(z)}{G(z)} dz = \sum_{a \in \mathfrak{m}_K} \text{res}_a \left(\frac{N}{G} dz \right), \quad (10.5)$$

a finite sum over actual poles, all of which lie strictly inside the chosen normalized circle. If $G = f + E$ with f of order m , there are exactly m denominator zeros counted with multiplicity, and the value agrees with (2.5).

Proof. If the leading germ is a unit then G is a unit of $\mathcal{A}_1$ and the assertion follows from theorem 10.2 or from coefficientwise primitives. Otherwise Hahn-germ preparation and division [2, §4] give $N/G = J + R/P$ with $J \in \mathcal{A}_1$, P monic of degree m with all roots infinitesimal, and $\deg R < m$, the normalization absorbing the leading monomial and unit factors into the numerator. These

roots are the zeros of G with multiplicities and satisfy the protecting bound of [lemma 10.3](#). Factor P over the algebraically closed K and use finite partial fractions $R/P = \sum_a \sum_{j=1}^{m_a} c_{a,j}(z-a)^{-j}$. The J integral is zero. For each a the normalized position a/r is infinitesimal, so expansion on $z = rw$, $|w| = 1$, is strongly summable, with normalized integral 1 for $(z-a)^{-1} dz$ and 0 for every $(z-a)^{-j} dz$ with $j > 1$. Hence the integral is $\sum_a c_{a,1}$. That no other infinitesimal zero exists is [lemma 10.3](#); agreement with (2.5) is [theorem 10.4](#), and the same finite set of local residues proves independence of ρ . $\square$

This extends the common-domain moving-pole theorem (8.10) to the radius-free setting without requiring a common ordinary representative for the original infinitely many coefficient germs. It is *not* a claim that the same germ can be evaluated at arbitrary noninfinitesimal points, nor that its original coefficient functions have acquired a common ordinary analytic radius. An arbitrary Hahn family of meromorphic germs is a larger object than a quotient of two elements of $\mathcal{A}_1$, and *we do not assert that every such family admits the microscopic contour construction*.

Example 10.6 (An explicit radius-free contour value). Take H from (2.2) and $F(z) = z$. For any $\rho > 0$, [theorem 10.4](#) gives

$$\frac{1}{2\pi i} \oint_{|z|=t^\rho}^H \frac{H(z)}{z} dz = H(0) = \sum_{j \geq 1} t^j = \frac{t}{1-t}. \quad (10.6)$$

After scaling the integrand is $\sum_{j \geq 1} \sum_{k \geq 0} j^k t^{j+k\rho} w^{k-1} dw$; at each exponent the coefficient is a finite Laurent polynomial and the residue selects exactly $k = 0$. This calculation is valid also when Γ has higher rank. The output is a strong sum, not a claim about convergence of its partial sums in any topology. With $\Gamma = \mathbb{Q}$, $a = t^2$ and $r = t^{1/2}$ the same mechanism gives

$$\frac{1}{2\pi i} \oint_{|z|=t^{1/2}}^H \frac{H(z)}{z-t^2} dz = \sum_{j \geq 1} \frac{t^j}{1-jt^2} = t + t^2 + 2t^3 + 3t^4 + 5t^5 + 9t^6 + 16t^7 + 31t^8 + \dots, \quad (10.7)$$

the coefficient at exponent ℓ being the finite sum of j^k over $j \geq 1$, $k \geq 0$ with $j + 2k = \ell$.

Example 10.7 (A radius-free numerator over a perturbed denominator). Take the numerator (2.2) and $F(z) = z^2 - t$, so $m = 2$ and $v_H(E) = 1$. For the protecting radius $r = t^{1/4}$, [theorem 10.4](#) gives

$$\begin{aligned} \frac{1}{2\pi i} \oint_{|z|=r}^H \frac{H(z)}{z^2-t} dz &= \sum_{k \geq 0} t^k [z^{2k+1}] H(z) \\ &= \sum_{j \geq 1} \sum_{k \geq 0} j^{2k+1} t^{j+k} = \sum_{j \geq 1} \frac{j t^j}{1-j^2 t} \\ &= t + 3t^2 + 12t^3 + 64t^4 + 441t^5 + 3855t^6 + \dots \end{aligned} \quad (10.8)$$

Every rearrangement is strongly valid: the coefficient at t^ℓ has only the pairs $j + k = \ell$ with $j \geq 1$, $k \geq 0$. Equivalently the actual simple-pole formula $(H(t^{1/2}) - H(-t^{1/2})) / (2t^{1/2})$ gives $jt^j / (1-j^2t)$ for each summand of H , proving agreement. *No common ordinary disk for the original numerator coefficients exists*, so the period is genuinely supplied by the microscopic representation and not by an unavailable fixed-circle integral.

11 Microscopic tori and general complete-intersection residues

11.1 Coordinate-power leading systems

Let

$$F_j(z) = z_j^{d_j} + E_j(z), \quad d_j \geq 1, \quad E_j \in \mathcal{A}_n, \quad v_H(E_j) = \delta_j > 0 \quad (11.1)$$

for each nonzero E_j (with $\delta_j = +\infty$ if $E_j = 0$). Choose positive $\rho_j \in \Gamma$ with

$$d_j \rho_j < \delta_j \quad \text{whenever } E_j \neq 0, \quad r_j = t^{\rho_j}; \quad (11.2)$$

such choices exist by divisibility of Γ , and for a zero perturbation there is no upper restriction on ρ_j . Write T_r for the parameterized real n -torus $z_j = r_j e^{i\theta_j}$, $0 \leq \theta_j \leq 2\pi$, with ordinary parameter angles and orientation $d\theta_1 \wedge \cdots \wedge d\theta_n$; integration is defined by pullback to the ordinary product of unit circles. *It is a real n -dimensional cycle in an n -complex-variable calculation, not the $(2n-1)$ -dimensional sphere used in some other several-variable integral formulas.*

Theorem 11.1 (Coordinate-power torus realization). *For F as in (11.1), radii as in (11.2), and $H \in \mathcal{A}_n$, the integral*

$$\mathcal{R}_F(H) := \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) dz_1 \wedge \cdots \wedge dz_n}{F_1(z) \cdots F_n(z)} \quad (11.3)$$

is well defined and satisfies the exact coefficient formula

$$\mathcal{R}_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} [z_1^{d_1(k_1+1)-1} \cdots z_n^{d_n(k_n+1)-1}] \left(H \prod_{j=1}^n E_j^{k_j} \right) = \text{Res}_F(H), \quad (11.4)$$

the perturbation residue (2.4) for $f_j = z_j^{d_j}$. Its support is contained in $\text{supp } H + E^$, where $E = \bigcup_j \text{supp } E_j$, and it is independent of all protecting radii satisfying (11.2). Every infinitesimal common zero satisfies $v(z_j) > \rho_j$ for every j .*

Proof. After substituting $z_j = r_j w_j$ one has $r_j^{-d_j} F_j(rw) = w_j^{d_j} + P_j(w)$ with $v_H(P_j) \geq \delta_j - d_j \rho_j > 0$, and all coefficient functions are polynomials by [theorem 10.1](#). On a common ordinary product annulus about $|w_j| = 1$ all leading $w_j^{d_j}$ are nonzero, so each reciprocal has a strong geometric expansion; [lemma 2.2](#) proves strong summability of the entire pulled-back form, which may therefore be integrated coefficientwise. At multi-index k the denominator contributes $\prod_j z_j^{-d_j(k_j+1)}$ and the numerator is $H \prod_j E_j^{k_j}$; the ordinary iterated torus integral selects the Laurent monomial $w_1^{-1} \cdots w_n^{-1}$, so a source Taylor monomial z^α contributes only when $\alpha_j = d_j(k_j+1) - 1$ for every j , and for that monomial the scale factor in coordinate j is $r_j^{1+\alpha_j-d_j(k_j+1)} = 1$. The surviving coefficients are exactly (11.4), including the sign. Ordinary Taylor coefficient operators do not enlarge Hahn support and the perturbations have positive support, giving the support statement; the formula contains no r_j , giving radius independence. Finally, for an infinitesimal common zero, $d_j v(z_j) = v(E_j(z)) \geq \delta_j$ whenever $z_j \neq 0$, and a zero coordinate has valuation $+\infty$. $\square$

The theorem permits *different valuation ranks in different coordinates*: for example $\rho_1 \in \mathbb{Q}_{>0}$ and ρ_2 proportional to ω are both legitimate, and nothing in the proof replaces Γ by a real-valued norm. It does not assume that the perturbations are polynomial, nor that their coefficient germs share an ordinary neighborhood. It supplies a contour meaning to a residue that was deliberately defined algebraically in [2].

Example 11.2 (A two-variable torus and cancellation across valuation ranks). Take $F_1 = x^2 - Ay$, $F_2 = y^2 - Bx$ with $A, B \in \mathfrak{m}_K \setminus \{0\}$, and choose $2\rho_x < v(A)$, $2\rho_y < v(B)$. Then

$$\mathcal{R}_F(H) = \sum_{k, \ell \geq 0} A^k B^\ell [x^{2k+1-\ell} y^{2\ell+1-k}] H, \quad (11.5)$$

a coefficient with a negative exponent being zero. In particular $\mathcal{R}_F(1) = 0$, $\mathcal{R}_F(xy) = 1$, and $\mathcal{R}_F(4xy - AB) = 4$. The Jacobian is $J_F = 4xy - AB$, equal to $-AB$ at the origin and to $3AB$ at the three roots with $xy = AB$, so the four weights for numerator 1 are $-1/(AB)$ and $1/(3AB)$ three times, summing to zero. For $A = t$ and $B = t^\omega$ this cancellation involves weights of valuation $-(1 + \omega)$, whereas the torus result has no negative term at all.

11.2 Arbitrary isolated analytic complete intersections

Now let $f = (f_1, \dots, f_n)$ be an ordinary analytic germ at zero with an isolated common zero, i.e. $0 < \mu = \dim_{\mathbb{C}} \mathbb{C}\{z\}/(f_1, \dots, f_n) < \infty$, and let $F = f + E$ with all nonzero $E_j \in \mathcal{A}_n$ of strictly positive Hahn valuation. The perturbation residue is (2.4), where the ordinary local Grothendieck residue is a complex-linear functional on analytic germs whose coordinate-power case is Taylor coefficient extraction. We use the classical residue transformation law as presented in [28, 9].

Since (f) has finite colength, choose $d_1, \dots, d_n \geq 1$ and an *ordinary analytic* matrix $A(z)$ with

$$A(z)f(z) = (z_1^{d_1}, \dots, z_n^{d_n})^\top, \quad (11.6)$$

and set $G = AF$, so that $G_j = z_j^{d_j} + D_j$ with $D = AE$ and all nonzero D_j of positive Hahn valuation. *No invertibility of A is assumed.* The existence of such A and d_j uses finite colength alone; it is ordinary local analytic algebra and requires nothing from the geometry manuscript beyond the definition of Res_F .

Lemma 11.3 (Transformation of the Hahn perturbation residue). *With these hypotheses, for every $H \in \mathcal{A}_n$,*

$$\text{Res}_F(H) = \text{Res}_G(H \det A). \quad (11.7)$$

Proof. We first specify the classical input. For a finite analytic parameter deformation f_u of an isolated complete intersection, the sum of local residues over the nearby cluster is analytic in u ; if $g_u = Af_u$ also has an isolated central fibre, the classical transformation law gives

$$\text{res}_{f_u}^{\text{cluster}}(h) = \text{res}_{g_u}^{\text{cluster}}(h \det A).$$

The cluster for g_u can contain additional points, and *the determinant is part of the law.* Taylor expansion in u produces the geometric denominator expansions appearing in (2.4). This ordinary finite-family statement follows from the usual local residue transformation law and its parameter version [28, 9].

Now fix a Hahn exponent ν . On either side of (11.7), a coefficient is built from a source coefficient of H together with a finite word of positive perturbation exponents. Lemma 2.2 leaves only finitely many contributing words, hence finitely many ordinary coefficient germs — this is exactly the place where the distinction recorded after example 2.3 is load-bearing, since infinitely many *exponents* may lie below ν . Replace the finitely many retained perturbation terms by independent ordinary parameters. Together with the finitely many entries of A , their germs have a common ordinary representative neighborhood. Apply the finite-family identity there, compare the finitely many

required Taylor coefficients, and substitute the monomials t^e back for the parameters; exponent collisions cause only finite additions. Thus the two sides have equal coefficient at ν , and ν was arbitrary. A fully detailed truncation argument for a transformation matrix that is only *formal* in the parameters is given in [section B](#). $\square$

Theorem 11.4 (Microscopic contour representation of a general residue). *Choose A and d_j as in (11.6) and protecting radii for G as in (11.2). Then for every $H \in \mathcal{A}_n$*

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) \det A(z) dz_1 \wedge \cdots \wedge dz_n}{G_1(z) \cdots G_n(z)}. \quad (11.8)$$

The value is independent of the admissible radii and of the choice of ordinary transformation data A, d_j .

Proof. The integrand is admissible by [theorem 10.1](#), and [theorem 11.1](#) identifies its normalized torus integral with $\text{Res}_G(H \det A)$. Apply [lemma 11.3](#). Every permitted choice gives the same independently defined left side, proving independence. $\square$

Two non-claims must survive this theorem, and are stated here as part of it. First, (11.8) is a contour representation *by a transformed form*. It does *not* assert that the untransformed form $H dz_1 \wedge \cdots \wedge dz_n / (F_1 \cdots F_n)$ has the same integral over that coordinate torus; in the classical theory the natural Grothendieck cycle is adapted to the equations, typically $\{|F_j| = \epsilon_j\}$, and replacing that cycle by a coordinate torus without the determinant transformation would in general be wrong. Second, the theorem does *not* claim that F and G have identical zero schemes: multiplication by a matrix with nonunit determinant can introduce additional zeros and multiplicities, and $\det A$ *compensates for this in the residue identity* — *it is not an optional Jacobian factor*. Formula (11.8) is useful precisely because it gives a controlled coordinate-torus substitute.

The logical cost of this theorem should be recorded exactly. It uses the strong residue expansion (2.4), ordinary finite-family residue transformation, and positive rescaling [theorem 10.1](#). *It does not require the Nullstellensatz or faithful-flatness theorems of [2], nor that manuscript's finite-algebra normal form.*

11.3 Relation to the finite-algebra conclusions of the supplied source

The geometry manuscript proves that the perturbed quotient $B_F = \mathcal{A}_n / (F_1, \dots, F_n)$ has K -dimension exactly μ , and that

$$\text{Res}_F(H J_F) = \text{Tr}(M_H) = \sum_a e_a H(a), \quad J_F = \det\left(\frac{\partial F_i}{\partial z_j}\right), \quad \sum_a e_a = \mu, \quad (11.9)$$

with Res_F annihilating (F) and e_a the local multiplicity at the *actual displaced* zero. For simple zeros this gives $\text{Res}_F(H) = \sum_a H(a) / J_F(a)$. These are reported companion results, not reproved here as part of the contour construction. Combining them with [theorem 11.4](#) identifies one functional simultaneously as a finite-algebra trace pairing and as a microscopic integral, and gives

$$\frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H J_F \det A}{G_1 \cdots G_n} dz_1 \wedge \cdots \wedge dz_n = \sum_a e_a H(a), \quad (11.10)$$

whose value at $H = 1$ is the ordinary integer μ . Thus this higher-dimensional contour has a finite local-degree interpretation even when the actual points split across several infinitesimal ranks. On a

common scaled neighborhood avoiding the divisor $G_1 \cdots G_n = 0$ the displayed meromorphic top form is holomorphic and closed, so its admissible deformations obey [theorem 7.4](#) and their periods agree when the intervening chain avoids that divisor — a genuine Stokes explanation of contour deformation in addition to the algebraic radius-independence proof. This is *not* a statement about arbitrary fine-continuous hypersurfaces in the full class plane.

In dimension one, Stokes, the residue of a Laurent series, and the integer winding of a circle already organize the calculation. In higher dimensions the determinant transformation and the real n -torus play the corresponding roles. *There is no need to postulate a new Jordan separation theorem for a $(2n - 1)$ -sphere in order to justify these residues.*

11.4 Scope: what a general isolated complete intersection still requires

This subsection records a disagreement about *status* among the three manuscripts merged here. It is the sharpest in the set, and it is not averaged. [Lemma 11.3](#) and [theorem 11.4](#) above are proved unconditionally and are printed with their proof. One of the three sources judged the corresponding statement *unproved* and declined to assert it, and a third reached a determinant torus formula only conditionally. Both positions are reproduced here, because dropping the objection would overstate how much is proved unconditionally, while dropping the theorem would understate it.

The objection, as raised. For arbitrary isolated f , a single diagonal rescaling need not make the leading equations coordinate powers, or even keep the leading homogeneous system isolated. Thus [theorem 11.1](#) cannot simply be applied after deleting its hypothesis. Moreover the usual cycle $\{|f_i| = \epsilon_i\}$ is not in general a coordinate product torus. What is available without further argument is an exact *coefficient-dependent cycle interpretation* of (2.4): at a prescribed Hahn exponent, [lemma 2.2](#) retains only finitely many numerator germs, perturbation germs, and multi-indices; those finitely many germs have a common ordinary representative neighborhood; their ordinary residues can be computed on suitable protecting cycles there; and the answer is independent of the representatives and radii by ordinary local residue theory. Doing this at each exponent gives precisely (2.4). *This is a compatible germwise procedure, not one fixed ordinary cycle for the entire original infinite family.* On that basis the source in question wrote: “The present article proves that realization for one variable and for the coordinate-power class above; it does not assume it for every complete intersection,” and listed a general realization theorem as an open problem: given arbitrary isolated f and radius-free positive-support E , construct *one* admissible n -cycle whose integral equals (2.4), possibly requiring a non-diagonal analytic change of variables, a finite resolution-type decomposition, or chains on an enlarged analytic space, and prove that no omitted summation or radius assumption is hidden in the parameterization.

What the objection does and does not refute. It does not refute [theorem 11.4](#), and the two are consistent, because they answer different questions. The open problem asks for a cycle on which the *untransformed* integrand $H dz/(F_1 \cdots F_n)$ is integrated; [theorem 11.4](#) supplies instead a coordinate torus for the *transformed* integrand $H \det A dz/(G_1 \cdots G_n)$, and, as stated immediately after that theorem, it explicitly does not claim the former. The objection is therefore best read as a statement about which of the two representations has been achieved. Nor does the objection’s existence prove that no single suitably nonstandard cycle is possible. Conversely, [theorem 11.4](#) does settle the representation question in the transformed form, unconditionally, from finite colength alone. A reader should take away both: a general isolated complete intersection *does* have a controlled microscopic contour representation, and it is *not* the naive one.

The conditional third route. A second and genuinely independent route to a determinant torus formula runs through univariate separating denominators. It is worth recording as an alternative proof, *provided its dependence is stated*: unlike [theorem 11.4](#), it assumes the finite-algebra and support-controlled normal-form results of the companion geometry manuscript [2], which is an unrefereed research manuscript. It is therefore conditional on that companion rather than independent of it.

Theorem 11.5 (Separating-torus realization; conditional on [2]). Assume the finite-algebra and support-controlled normal-form results of [2, §§7–9]: that $B_F = \mathcal{A}_n/(F_1, \dots, F_n)$ has K -dimension μ , that any chosen ordinary monomial basis remains a basis, and that multiplication matrices and normal-form operations have explicit support in E^* . Let M_i be the multiplication matrix of z_i on B_F in that basis, and put

$$Q_i(T) = \det(TI - M_i) = T^\mu + \sum_{j < \mu} q_{ij} T^j. \quad (11.11)$$

Its exponent-zero matrix is multiplication by z_i in the ordinary local algebra $\mathbb{C}\{z\}/(f)$, hence nilpotent, so every nonzero q_{ij} has positive valuation. By Cayley–Hamilton $Q_i(z_i) \in (F)$; applying the supplied normal form to it, with zero remainder, produces a matrix $C = (C_{ij})$ with

$$Q_i(z_i) = \sum_{j=1}^n C_{ij}(z) F_j(z), \quad C_{ij} \in \mathcal{A}_n, \quad (11.12)$$

all entries having nonnegative support contained in E^* . Choose $\rho_i \in \Gamma_{>0}$ with

$$(\mu - j)\rho_i < v(q_{ij}) \text{ for every nonzero } q_{ij}, \quad r_i = t^{\rho_i}. \quad (11.13)$$

Then for every $H \in \mathcal{A}_n$

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^n} \int_{T_r}^H \frac{H(z) \det C(z)}{\prod_{i=1}^n Q_i(z_i)} dz_1 \wedge \dots \wedge dz_n, \quad (11.14)$$

independently of the allowed radii and of the chosen representing matrix C , with support contained in $\text{supp } H + E^*$.

Proof. Existence of the integral. By [theorem 10.1](#), $H(rw)$ and every entry of $C(rw)$ have polynomial complex coefficients at each Hahn exponent. The scaled denominator is $r_i^{-\mu} Q_i(r_i w_i) = w_i^\mu + \sum_{j < \mu} q_{ij} r_i^{j-\mu} w_i^j$, whose non-leading coefficients have strictly positive valuation by (11.13); hence each reciprocal is a strong geometric expansion near the ordinary product unit torus, and products with the differential factors $r_i dw_i$ remain strongly supported. Moreover every root b of Q_i satisfies $v(b) > \rho_i$, since otherwise b^μ has strictly smaller valuation than each $q_{ij} b^j$ and cannot cancel; so every common zero of F lies strictly inside the separating coordinate circles.

Residue transformation. The classical rule $\text{Res}_F(H) = \text{Res}_Q(H \det C)$ for $Q = CF$ is standard residue theory [9, §0]. Its justification here is the finite-parameter argument of [section B](#), which is needed because C arises from a linear splitting that need not be continuous in an analytic norm, so $C(u)$ cannot be asserted convergent as an analytic parameter series.

The separated residue is a torus integral. For the leading tuple $(z_1^\mu, \dots, z_n^\mu)$ the ordinary residues are coefficient extraction; expanding each Q_i^{-1} around $z_i^{-\mu}$ shows that Res_Q is the iterated coefficient of $z_1^{-1} \dots z_n^{-1}$, and after the admissible scaling, integration of an ordinary Laurent monomial on each unit circle extracts exactly those coefficients, the powers of r_i cancelling. All rearrangements are strongly summable by the verified support bounds. The left side is already independent of the radii and of C , so equality gives the remaining assertions; no uniqueness of C is needed. $\square$

Remark 11.6 (What was changed, and what was not). Formula (11.14) does *not* assert that the original quotient $H/(F_1 \cdots F_n)$ is regular on the chosen product torus: a hypersurface $F_i = 0$ can meet that torus. The denominators were deliberately replaced by the univariate Q_i and $\det C$ was inserted. *Omitting the determinant is generally wrong.* Nor has a real n -torus been identified with the full boundary of a complex polydisk; the integration cycle and its real dimension are exactly those specified. Using univariate polynomials in a zero-dimensional ideal to compute multidimensional residues is a classical method [9, §0]; the additional conclusion here is that the method admits a legitimate support-controlled contour interpretation for $\mathcal{A}_n$. No claim is made that separating denominators or residue transformation are new ideas.

Which representation is sharper, and for what. Theorem 11.4 is sharper for *logical economy*: it is unconditional, needing only finite colength, ordinary residue transformation, and theorem 10.1, so it survives if the companion manuscript's finite-algebra package is withdrawn. Theorem 11.5 is sharper for *computation and for zero-scheme bookkeeping*: its denominators are explicit univariate characteristic polynomials whose roots and admissible radii are read off from (11.13), and the vanishing of $\det C$ at the spurious points of the separated system is visible, as example 11.7 shows. They are not competing proofs of one statement; they certify different things.

Example 11.7 (A four-point cluster and its separating torus). Take $F_1 = x^2 - Ay$, $F_2 = y^2 - Bx$ with $A, B \in \mathbf{m}_K \setminus \{0\}$ as in example 11.2. The ordinary leading algebra is $\mathbb{C}\{x, y\}/(x^2, y^2)$ with basis $1, x, y, xy$ and $\mu = 4$. The coordinate characteristic polynomials are

$$Q_x = x^4 - A^2 Bx, \quad Q_y = y^4 - AB^2 y,$$

and an explicit transformation matrix with $(Q_x, Q_y)^\top = C(F_1, F_2)^\top$ is

$$C = \begin{pmatrix} x^2 + Ay & A^2 \\ B^2 & y^2 + Bx \end{pmatrix}, \quad D = \det C = x^2 y^2 + Bx^3 + Ay^3 + ABxy - A^2 B^2. \quad (11.15)$$

Hence

$$\text{Res}_F(H) = \frac{1}{(2\pi i)^2} \int_{T_r}^H \frac{H(x, y) D(x, y)}{(x^4 - A^2 Bx)(y^4 - AB^2 y)} dx \wedge dy, \quad (11.16)$$

admissible for any positive exponents with $3\rho_x < 2v(A) + v(B)$ and $3\rho_y < v(A) + 2v(B)$; with $A = t$ and $B = t^\omega$ both $\rho_x = \rho_y = 1/4$ work. The roots are $(0, 0)$ and $(\varrho\zeta^j, \varrho^2\zeta^{2j}/A)$ for $j = 0, 1, 2$ with $\varrho^3 = A^2 B$ and ζ a primitive cube root of unity. The weights $-1/(AB)$, $1/(3AB)$, $1/(3AB)$, $1/(3AB)$ are individually infinite and sum to zero, while $\text{Res}_F(xy) = 1$ and $\text{Res}_F(J_F) = 4$. In the quotient algebra $D = ABJ_F$, the exact polynomial identity having a remainder in (F_1, F_2) .

The determinant's role is especially explicit here. *The separated system (Q_x, Q_y) has sixteen simple grid points, whereas the original system has only four.* At each of the other twelve points $CF = 0$ with $F \neq 0$, so $\det C = 0$ and the separated simple residue vanishes. Changing the denominators without the determinant would count the wrong zero scheme. For an independent check, expanding $Q_x^{-1} = x^{-4} \sum_{k \geq 0} (A^2 B)^k x^{-3k}$ and $Q_y^{-1} = y^{-4} \sum_{\ell \geq 0} (AB^2)^\ell y^{-3\ell}$ on the chosen torus and extracting the coefficient of $x^{-1}y^{-1}$ reproduces

$$\text{Res}_F(x^p y^q) = \begin{cases} A^{(2p+q-3)/3} B^{(p+2q-3)/3}, & \frac{2p+q-3}{3}, \frac{p+2q-3}{3} \in \mathbb{N}, \\ 0, & \text{otherwise,} \end{cases} \quad (11.17)$$

in agreement with (11.5). The general equality is proved by theorem 11.5, not inferred from the finite tests.

12 An algebraic contour pairing for rational differentials

The Hahn constructions above rely on an ordinary base or a useful infinitesimal chart. Rational functions admit a different, fully algebraic closed-period theory over any fixed real closed workspace, requiring neither an infinite sum nor a limiting partition. In this section the coefficient ring is the rational function field $K(z)$ over $K = R[i]$, R real closed containing $\mathbb{R}$; it is none of the four rings of [section 2.3](#), and no statement below is transferred to or from them except through the explicitly proved comparison [proposition 12.4](#).

12.1 Definition from definable winding

Let $\gamma : [0, 1]_R \rightarrow K$ be a semialgebraic continuous loop, piecewise smooth when differential notation is used, or more generally a finite oriented semialgebraic cycle. For $a \notin \gamma$ write $\text{wind}_{\text{sa}}(\gamma, a) \in \mathbb{Z}$ for its semialgebraic winding number as in [section 4](#), normalized to 1 for a positively oriented circle about an inside point and 0 outside.

Definition 12.1 (Algebraic rational contour functional). For a rational differential $\omega = Q(z) dz$, $Q \in K(z)$, with no pole on γ , define

$$I_{\text{alg}}(\gamma, \omega) := 2\pi i \sum_{a \in K} \text{wind}_{\text{sa}}(\gamma, a) \text{res}_a \omega. \quad (12.1)$$

Only the finitely many finite poles contribute, so the sum is finite. As fixed in [section 2.4](#), the factor $2\pi i$ is a normalization selected to match ordinary complex integration using the distinguished copy of $\mathbb{C}$ inside K ; *it is not forced by ordered-field arithmetic*, and over an abstract real closed field with no distinguished copy of $\mathbb{R}$ one would normalize formally or use $I/(2\pi i)$. A residue at infinity could be included after compactifying and changing charts, but is unnecessary for this affine formula and supplies no new generator, its residue being the negative sum of the finite ones.

Equation (12.1) is the definition of a pairing. It is not a residue theorem deduced from a previously constructed Riemann integral, and it is not a claim that a finite-partition Riemann net converges to this value. Its ingredients are algebraic local residues and the independently defined definable degree. It is useful because it requires neither ordinary leading separation nor a global logarithm: poles and contour coefficients may lie at arbitrary Hahn scales.

Theorem 12.2 (Characterization and algebraic Cauchy theory). *For a fixed semialgebraic loop γ , (12.1) is the unique K -linear pairing on rational differentials without poles on γ which vanishes on exact rational differentials and satisfies*

$$I_{\text{alg}}\left(\gamma, \frac{dz}{z-a}\right) = 2\pi i \text{wind}_{\text{sa}}(\gamma, a).$$

It is additive under concatenation of loops and additive in cycles, changes sign under orientation reversal, and is invariant under definable homotopies avoiding the poles. If all indices at the poles vanish, the integral is zero. If γ is a positively oriented semialgebraic Jordan curve and Q has no poles on its bounded side or on its boundary, then $I_{\text{alg}}(\gamma, Q dz) = 0$; and if a lies on that bounded side, then

$$I_{\text{alg}}\left(\gamma, \frac{Q(z)}{z-a} dz\right) = 2\pi i Q(a).$$

Finally, for a nonzero $Q \in K(z)$ with neither zeros nor poles on γ ,

$$\frac{1}{2\pi i} I_{\text{alg}}\left(\gamma, \frac{Q'}{Q} dz\right) = \sum_a \text{wind}_{\text{sa}}(\gamma, a) \text{ord}_a(Q), \quad (12.2)$$

negative orders counting poles.

Proof. Since K is algebraically closed of characteristic zero, partial fractions give $Q(z) = P(z) + \sum_a \sum_{j=1}^{m_a} c_{a,j}(z-a)^{-j}$. The polynomial part has a polynomial primitive, and for $j \geq 2$

$$\frac{dz}{(z-a)^j} = d\left(\frac{(z-a)^{1-j}}{1-j}\right).$$

Thus a pairing with the stated properties must retain exactly the $j = 1$ terms with the displayed winding coefficients; this proves uniqueness and simultaneously verifies that (12.1) has those properties, since the residue of a rational derivative is zero (differentiating a Laurent term cannot produce exponent -1). Additivity, orientation behaviour and homotopy invariance follow from the corresponding properties of wind_{sa} . For a positively oriented Jordan curve, exterior poles have winding zero; in the Cauchy-kernel expression the only contributing interior pole is a , with residue $Q(a)$. If $Q = (z-a)^m V$ with $V(a) \neq 0$, then $Q'/Q = m/(z-a) + V'/V$ has residue m at a , giving (12.2). $\square$

Equation (12.2) is a rational argument principle over the full real closed workspace; for a simple positively oriented Jordan cycle its right side is the number of inside zeros minus inside poles, with multiplicities, an exact arbitrary-scale counterpart of the classical statement. Eisermann's polynomial Cauchy and root-counting theorems provide the effective winding component for polynomially described contours [15, 25], subject to the effectivity caveat of section 4.

12.2 Residues are the complete rational obstruction

Proposition 12.3 (Rational de Rham quotient). *Over the algebraically closed characteristic-zero field K ,*

$$\frac{K(z) dz}{dK(z)} \simeq \bigoplus_{a \in K} K\left[\frac{dz}{z-a}\right], \quad (12.3)$$

and the coordinates of a class are its residues at the finite points.

Proof. Partial fractions express every rational differential as a polynomial differential plus finitely many pole terms. Polynomial terms integrate by division by positive integers; for $j \geq 2$, $(z-a)^{-j} dz$ is the derivative of $-(z-a)^{-j+1}/(j-1)$. Only simple poles remain. The residue of a derivative is zero, so a nonzero linear combination of distinct simple-pole terms cannot be exact. This proves both spanning and independence. $\square$

Consequently (12.1) is the winding pairing with the *entire* rational de Rham quotient, not an arbitrary selection of part of the rational information. The corresponding local statement is purely formal: for $\omega = \sum_{j \geq j_0} c_j u^j du \in K((u)) du$ there is a formal Laurent primitive exactly when $c_{-1} = 0$, since every other term integrates by division by $j+1$ while differentiating a Laurent series can never produce a u^{-1} coefficient. Hence

$$K((u)) du / dK((u)) \simeq K, \quad [\omega] \longmapsto c_{-1}, \quad (12.4)$$

and the residue is invariant under an invertible formal coordinate change $u = v h(v)$, $h(0) \neq 0$: exact differentials remain exact, and $du/u = dv/v + h'(v) dv/h(v)$, whose second term has zero residue. Residues therefore give a coordinate-invariant obstruction to exactness independently of any connected topology. This also explains the difference from the fine-local category: a global fine-analytic primitive of $1/z$ may exist after a phase convention, but no rational primitive exists, and the cohomology computed in (12.3) is that of a differential algebra, not unrestricted fine-local cohomology.

Finite rational and semialgebraic data over **SC** can always be placed in a set-sized real closed workspace: the relevant roots lie in its algebraic closure K , and enlarging the workspace changes neither finite sign computations nor the residue formula. This gives an especially robust full-surcomplex interpretation for finite algebraic input.

12.3 Agreement with Hahn integration on a verified overlap

Proposition 12.4 (Comparison on separated charts). *Suppose a semialgebraic loop γ has an affine normalized description $C = a + r\gamma_R$, where γ is an ordinary piecewise-polynomial closed loop and γ_R its semialgebraic extension to field-valued parameters, whose restriction to ordinary parameters is an admissible Hahn-deformed contour $\gamma_0 + \eta$. Assume that the same description on the full field interval is uniformly infinitesimally close to its ordinary real-coefficient leading loop. Suppose each pole b of $\omega = Q dz$, $Q \in K(z)$, has either finite normalized position $(b - a)/r$ with standard part off the ordinary trace of γ_0 , or infinite normalized position. Then, whenever the rational pulled-back differential is a coherent family on a common ordinary collar,*

$$I_{\text{alg}}(\gamma, \omega) = \int_{\gamma}^H \omega.$$

The same holds after a polynomial infinitesimal deformation for which a definable pole-avoiding homotopy to this loop is verified.

Proof. Use finite partial fractions after the affine substitution. Polynomial terms have coherent primitives and zero period; a term of order greater than one has a rational primitive, coherent on the coefficient collar, and also zero period. For a finite pole b with normalized position $c + \varepsilon$, ordinary compactness gives a positive ordinary separation between γ_0 and c ; the real-coefficient semialgebraic description has the same lower bound on the field interval, and uniform infinitesimal closeness makes the straight definable homotopy from (γ, b) to (γ_0, c) avoid zero differences throughout, so $\text{wind}_{\text{sa}}(\gamma, b) = \text{wind}(\gamma_0, c)$. Theorem 8.4 then gives equality of the two pairings on $dz/(z - b)$. If the normalized pole is infinite, its inverse is infinitesimal, the reciprocal has a coherent expansion in nonnegative powers of the finite chart variable with zero period, and the field-valued polynomial loop lies in the bounded part of that chart so the pole has index zero. Exact rational differentials have zero period by the coefficientwise fundamental theorem on the admitted pullbacks. Partial fractions now prove the assertion term by term, the affine scale factors cancelling as in (7.6). For the last sentence, use the specified definable homotopy for algebraic indices and theorems 7.5 and 8.2 for the coherent integrals. $\square$

The explicit overlap hypotheses prevent an unwarranted claim that every smooth Hahn-deformed path extends to a semialgebraic map on $[0, 1]_{R_F}$; most do not. The proposition says the two theories agree when their objects and their homotopies really can be compared. Agreement of the two constructions is *proved where both are defined*, not postulated across different chain categories.

[Propositions 9.1](#) and [12.4](#) are two separate compatibility results and must not be conflated. The first compares two *chart descriptions of one chain* within the Hahn category; the second compares *two different integration theories*. Here is the explicit case where conflating them fails. For the full semialgebraic unit circle, [\(12.1\)](#) gives immediately

$$I_{\text{alg}}\left(S^1, \frac{dz}{z - (1 - t)}\right) = 2\pi i, \quad I_{\text{alg}}\left(S^1, \frac{dz}{z - (1 + t)}\right) = 0. \quad (12.5)$$

These are legitimate values *in exactly the case where the unmodified macroscopic Hahn pullback fails*, namely the boundary-layer situation of [section 3.6](#) and [example 4.2](#). They do not contradict [proposition 12.4](#), whose leading-separation hypothesis is absent here; and they are not obtainable from [proposition 9.1](#), which presupposes admissible pullbacks that do not exist in this case.

12.4 What this pairing does not determine

For an *open* path, the differential $dz/(z - a)$ requires a logarithmic increment, not just a residue and a winding integer; a choice of path continuation or a more elaborate period object is needed, and a global fine logarithm is not automatically that choice by [section 3](#). For nonrational analytic functions, partial fractions need not leave a rational exact remainder, so [\(12.1\)](#) is *not* a definition on all of $\mathcal{H}(U)$ or $\mathcal{A}_n$; those larger classes are handled by the coherent and microscopic constructions, not by pretending that the rational formula already covers them. The functional also does not by itself extend to a general Hahn-holomorphic function with an infinite collection of coefficient singularities.

13 Root-of-unity sampling: a positive theorem and its indispensable hypothesis

Two of the merged manuscripts leave opposite impressions about root-of-unity quadrature: one presents it as a negative result, the other devotes a section to its positive side. They are compatible, and the hypothesis is doing all the work. This section therefore states the positive theorem *with* that hypothesis and keeps the negative example immediately alongside as the demonstration that the hypothesis cannot be dropped.

13.1 What the classical construction assumes

For a complete algebraically closed field with a real-valued non-Archimedean absolute value, a useful circle integral is a limit of the weighted samples

$$S_N(f; a, r) = \frac{r}{N} \sum_{\xi^N=1} f(a + r\xi) \xi, \quad (13.1)$$

the ξ being the ordinary complex roots of unity. Cherry's presentation of the Schnirelman integral uses this normalization, takes $N \rightarrow \infty$ with $|N| = 1$, and defines the integral only when the field-valued limit exists [[11](#), §1.3]; its residue normalization is 1, not $2\pi i$, and the chosen r is part of the datum for arbitrary functions, not just its absolute value. In a Hahn field with coefficient field $\mathbb{C}$ the natural valuation has residue characteristic zero and $v(N) = 0$ for every positive integer. Related averages appear in p -adic line integral theories [[14](#), [16](#)].

The basic algebraic fact is exact and requires no limit. If $f(a + w) = \sum_{j=j_0}^{j_1} c_j w^j$ is a Laurent polynomial, then

$$S_N(f; a, r) = \sum_{\substack{j_0 \leq j \leq j_1 \\ N \mid j+1}} c_j r^{j+1}, \quad (13.2)$$

from $N^{-1} \sum_{\xi^{N=1}} \xi^q = 1$ when $N \mid q$ and 0 otherwise. When N exceeds every nonzero $|j + 1|$ in the finite support, the average equals c_{-1} , which is exactly the corresponding normalized contour integral. *This is exact algebraic quadrature for finite Laurent data; it is not a convergence theorem.* The words “complete Hahn field” alone do not justify a limit: a common ordinary analytic coefficient expansion and a convergent non-Archimedean Laurent expansion are different hypotheses.

13.2 A rank-independent sufficient condition for a valuation limit

Proposition 13.1 (A precise sampling criterion). *Suppose $A_k \in K$, $k \in \mathbb{Z}$, satisfy the decay condition*

$$\text{for every } \gamma \in \Gamma, \quad \{k \in \mathbb{Z} : v(A_k) \leq \gamma\} \text{ is finite.} \quad (13.3)$$

Then $f(\xi) = \sum_{k \in \mathbb{Z}} A_k \xi^k$ is strongly defined at every root of unity, and

$$S_N(f; 0, 1) = \sum_{\ell \in \mathbb{Z}} A_{\ell N-1} \longrightarrow A_{-1} \quad (13.4)$$

in the valuation topology as the ordinary integer N tends to infinity. This holds for Γ of arbitrary rank.

Proof. For a fixed exponent bound γ , only finitely many A_k can contribute below that bound; their supports form a finite union of well-ordered sets. This proves that the full union is well ordered — a descending sequence would be bounded above by its first term and hence lie in that finite union — and also proves local finiteness of contributions. Multiplication by a root of unity does not change support, so the series defining $f(\xi)$ is strong. The finite root-of-unity sum gives $N^{-1} \sum_{\xi^{N=1}} \xi^{k+1} = 1$ when $N \mid k + 1$ and 0 otherwise, and strong summability permits applying this finite linear operation, proving the alias formula. Given γ , choose N_0 so large that the finitely many $k \neq -1$ with $v(A_k) \leq \gamma$ all satisfy $|k + 1| < N_0$; for $N \geq N_0$ none of them has $N \mid k + 1$, so every remaining nonprincipal term in (13.4) has valuation greater than γ , and so does the error. This is exactly valuation convergence. $\square$

The hypothesis is doing all the work, and its strength should be stated plainly. Condition (13.3) demands eventual domination of *every* value-group threshold; it is strictly stronger than the existence of a strong sum. Moreover, when no countable subset is cofinal in the positive value group, a nontrivial infinite sequence of finite valuations *cannot* satisfy this tail condition. This is one reason sequential constructions have a limited role at very large ranks.

13.3 Coefficientwise recovery always has a different meaning

Let $F = \sum_{\lambda \in S} f_\lambda t^\lambda$ have one well-ordered support and ordinary continuous coefficient functions on the unit circle. In the fixed support space $\mathbb{C}^S$ use the product topology of the ordinary complex coefficient topologies; this is a topology on *coefficient presentations* and is not the natural valuation topology of K .

Theorem 13.2 (Coefficientwise recovery of a Hahn contour). *The samples $S_N(F; 0, 1)$ have support contained in S and*

$$\lim_{N \rightarrow \infty}^{\text{coefficientwise}} S_N(F; 0, 1) = \frac{1}{2\pi i} \oint_{|z|=1}^H F(z) dz, \quad (13.5)$$

the right side being a Hahn series with support contained in S . If the f_λ are holomorphic on a common ordinary annulus — that is, if $F \in \mathcal{H}$ of that annulus — the coefficient at λ is their ordinary Laurent residue.

Proof. At exponent λ the sample is $N^{-1} \sum_{j=0}^{N-1} f_\lambda(e^{2\pi i j/N}) e^{2\pi i j/N}$, an ordinary Riemann sum for $(2\pi)^{-1} \int_0^{2\pi} f_\lambda(e^{i\theta}) e^{i\theta} d\theta$, which is the normalized ordinary contour integral; its convergence requires no uniformity in λ . Taking these limits creates no exponents outside the fixed well-ordered S , so the result is a legitimate element of K . The annulus assertion is the ordinary Laurent coefficient formula. $\square$

This realizes the already defined coefficientwise integral by finite samples in a *specified coefficient topology*. It does not make that integral a fine limit, and in general it does not make it a valuation limit either. Taking ordinary coefficient limits is a legitimate alternative description of the Hahn contour functional; it is not a limit in the internal valuation topology.

13.4 Why the hypothesis cannot be dropped

Example 13.3 (A failed valuation-limit contour definition). For $a = 0$, $r = 1$ and $f(z) = 1/(z - b)$ with ordinary real $b = 1/2$, logarithmic differentiation of $X^N - 1$ gives

$$S_N(f; 0, 1) = \frac{1}{N} \sum_{\xi^N=1} \frac{\xi}{\xi - b} = \frac{1}{1 - b^N}.$$

The ordinary complex limit is 1, but $v(S_N(f) - 1) = 0$ for every N , so there is no convergence to 1 in the Hahn valuation topology. In the full fine topology it cannot converge at all, since it is not eventually constant. If instead $b = t^\rho$ with $\rho > 0$, then $v(S_N(f) - 1) = N\rho$: a rank-one valuation can see convergence, but a higher-rank group containing an element larger than every $N\rho$ still cannot.

Example 13.4 (Ordinary convergence invisible to the Hahn valuation). Take the ordinary $f(z) = 1/(1 - z/2)$ on a neighborhood of the unit disk, embedded as the sole coefficient at exponent zero. Then

$$S_N(f; 0, 1) = \frac{2}{2^N - 1}. \quad (13.6)$$

It tends to 0 ordinarily and coefficientwise, agreeing with its zero contour integral, but every displayed nonzero number has Hahn valuation 0 and distinct samples differ by nonzero real amounts of valuation 0, so the sequence is not valuation Cauchy. Its ordinary Taylor coefficients 2^{-k} do not tend to zero in the natural non-Archimedean valuation. *No classical Schnirelman theorem on valuation-convergent analytic series is being contradicted.*

Example 13.5 (An obstruction occurring only in higher rank). In $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ let $f(z) = 1/(1 - tz)$. Its coherent unit-circle integral is zero, but

$$S_N(f; 0, 1) = \frac{t^{N-1}}{1 - t^N}, \quad v(S_N) = N - 1 < \omega. \quad (13.7)$$

The samples fail to converge to zero in the valuation topology, although each fixed Hahn coefficient is eventually zero. In a rank-one workspace with value group $\mathbb{Q}$ the same formula does converge to zero. In the full fine topology it cannot converge, since it is never eventually zero. *One exact formula therefore separates three notions of convergence.*

Both sample identities follow without any infinite-series argument: from $\prod_{\xi=1}^N (1 - q\xi) = 1 - q^N$, logarithmic differentiation gives $N^{-1} \sum_{\xi=1}^N \xi / (1 - q\xi) = q^{N-1} / (1 - q^N)$; substitute $q = 1/2$ and $q = t$. This calculation is also an effective diagnostic: before interpreting root-of-unity samples as an integral in a chosen topology, check whether their error valuations dominate the thresholds that topology actually uses. Finite Laurent data give exact algebraic quadrature; general coherent coefficients give ordinary coefficient limits; valuation convergence needs the additional cofinality and regularity hypothesis (13.3).

14 Other established approaches and their exact scope

The following theories are genuine precedents or alternatives. They do not all integrate the same forms, take values in the same field, or use the same notion of path. The comparisons are intentionally narrower than a claim of equivalence, and importing a theorem name without its hypotheses would be misleading.

14.1 Semialgebraic chains, Stokes, and the target of integration

Ohmoto and Shiota prove a C^1 triangulation theorem for semialgebraic sets and formulate integration and Stokes using finite differentiable simplices [23, Thm. 4.1, Prop. 4.2]. This is an important existing counterpart, not merely a proposal. Their non-Archimedean discussion explicitly warns that integral values need not remain in the given field and that a larger saturated target may enter; it should not be read as a construction of the particular K -valued contour operator defined here.

There is a further codomain distinction in the cited literature. Maříková and Shiota construct a measure with values in an ordered semiring, a Dedekind completion of a quotient, with a volume-preserving change of variables [21]; Berarducci and Otero’s earlier additive measure is real-valued on definable subsets of the finite part [7]; Kaiser develops semialgebraic measure and integration over non-Archimedean real closed fields containing $\mathbb{R}$ with *Archimedean* value group, using valuation-section data, and his construction can enlarge the value ring — for Puiseux series a polynomial ring in an additional variable occurs [18]. These are different targets retaining different information, and *a semiring-valued measure is not automatically a signed K -valued complex line integral*. A reference to any of them is not a proof that every signed or complex differential form has a canonical integral in the original Hahn field. Kaiser’s setting also warns that integrals of rational functions may require logarithmic values outside the initial workspace — which the polynomial theorem [theorem 5.1](#) avoids by using finite factorial identities, and which [theorem 12.2](#) avoids by using residues rather than open-path logarithms.

For a surcomplex application one must therefore specify the scalar integration operator on each simplex, its target, and an embedding or comparison map if values in $\mathbf{SC}$ are wanted. *Triangulation supplies a finite chain decomposition; it does not by itself solve the scalar integration problem.* In particular, saturation alone does not make an ordinary sequence $1/N$ converge to zero in the full ordered topology of a field containing infinitesimals, so the finite-mesh obstruction [proposition 3.5](#) remains unless the limiting notion has been changed. A promising combination is to use definable

triangulation to organize domains and then impose Hahn-coherent pullback conditions on each simplex; then [theorems 6.1](#) and [7.4](#) give Stokes whenever those checks succeed. *A theorem that all desired definable data admit such a finite coherent atlas would be an additional result, not a consequence of triangulation alone.*

14.2 Berkovich geometry changes the space, not just the topology

For a rank-one subgroup $\Gamma \subset \mathbb{R}$ put $|x|_v = \exp(-v(x))$. The Hahn field is then complete and algebraically closed, hence a legitimate base for Berkovich geometry. The Berkovich line adds multiplicative seminorms on $K[T]$, not merely more field points; it is tree-like and uniquely arc-connected, and deleting a branch point can create infinitely many components [[27](#), §2.3.3], [[8](#)]. Thus the enriched line acquires meaningful paths but *not* a planar Jordan circle: two distinct arcs around a circle would contradict uniqueness, and an annulus has an interval skeleton rather than a circular one. The appropriate boundary intuition is a disk point with outgoing directions, or the ends of an annular skeleton, not a fine-connected curve in K .

The rank restriction is substantive. An arbitrary higher-rank Γ cannot be embedded order-preservingly into $\mathbb{R}$; passing to a real-valued coarsening may lose infinitesimal scales, while retaining them requires a different higher-rank framework, and [example 9.3](#) exhibits distinctions that no single real-valued valuation can retain. Also $|c|_v = 1$ for every nonzero ordinary complex constant c , however small its usual complex modulus, so ordinary coefficient convergence is not non-Archimedean convergence. *No canonical reduction of all surreal scales to one rank is asserted here.*

14.3 Tropical differential forms and Stokes on Berkovich spaces

Chambert-Loir and Ducros construct real (p, q) -forms and currents on Berkovich spaces using tropicalizations and calibrated skeleta, with integration of top-bidegree forms and a Stokes formula; the expanded 2025 revision develops this in a broad tropical setting and includes a tropical Stokes formula [[12](#), Prop. 5.11.1]. This is a powerful established analogue of part of complex differential geometry, and a reason not to regard $d^2 = 0$ as an integration theorem by itself.

Its integrals are real-valued and its geometric objects are enriched analytic spaces. The K -valued periods of $F(z) dz$ constructed here, with the chosen normalization $2\pi i$, are different objects. A comparison could associate tropical quantities to valuations or norms of analytic data, but one should not expect that operation alone to recover complex phase or the cancellation of opposite residues such as the $\pm 1/(2t)$ of [example 8.8](#). A theorem connecting the pairings would need additional hypotheses and an explicit map on both forms and chains. *A Berkovich Stokes theorem cannot be quoted as the proof of the specific contour identities in the supplied manuscripts.*

14.4 Perfectoid analytic cycles: an actual Stokes theorem, but a different base

Mihara constructs analytic singular homology from a cosimplicial perfectoid object and integration over a restricted class of integrable analytic simplices, with [[20](#), Thm. 5.15] proving a Stokes formula for those simplices; in the integration part of the current text the base field is a finite extension of $\mathbb{Q}_p$ and the target is the de Rham period ring B_{dR} . This is directly relevant as evidence that replacing ordinary simplices and carefully restricting integrability can produce a non-Archimedean cycle pairing.

It is *not* a theorem for the present natural Hahn valuation with residue field $\mathbb{C}$: the residue characteristic and the integration target both differ, and the positive-residue-characteristic hypothesis

is decisive. Citing an older title or an abstract about “non-Archimedean integration” would not justify silently applying its period-ring construction to all surcomplex numbers. The common lesson is methodological: a chain category and its face compatibility must be constructed before a Stokes assertion is meaningful.

14.5 Surreal antiderivatives, resurgent extensions, and nonstandard bridges

Costin and Ehrlich construct extensions and integration for substantial classes of resurgent real functions, with treatments of singularities and ordered exponential subfields [10]. Their framework is a natural candidate when a contour is parameterized by functions whose real and imaginary pullbacks belong to those classes, and it can retain asymptotic information that a mere standard-part integral would lose. It is not merely the coefficientwise integral on a fixed ordinary domain.

To turn that into a contour theorem, however, one must prove closure under the required compositions $F(\gamma(s))\gamma'(s)$, specify logarithmic branches and endpoint conventions, and establish substitution and homotopy theorems; agreement with Hahn integration should first be tested on common analytic germs, rational forms, and the microscopic residues above. *Extending a real one-variable integral to surreal endpoints is not automatically a construction of complex contour integration*: complexification, phase and branch data, pullback along a path, and compatibility with a two-dimensional boundary operator remain to be established. Outside such an overlap, equality of integration operators should be a theorem, not an assumption. There is no conflict between the breadth of these function-extension theories and the arbitrary smooth coefficient forms used on ordinary parameter manifolds in section 6: the latter construction integrates ordinary functions first and stores their values as Hahn coefficients, and does not assert a canonical fine-smooth extension of every ordinary smooth function to every surcomplex point.

Ducros, Hrushovski, and Loeser construct a comparison between complex and non-Archimedean integrals using nonstandard models and compatible differential-form structures [13]. Their work shows that rigorous bridges exist, and also how much additional structure a bridge requires. Real closedness alone is not a transfer principle for arbitrary analytic functions, manifolds, or hyperfinite sums: a nonstandard model with a specified star map can transfer ordinary smooth integration and use internal hyperfinite sums, but the transferred domain, functions, and partitions must be internal, and standard-part or summation operations then determine what information the output retains. *It is invalid to call a proper-class surreal interval an internal hyperfinite interval merely because both contain infinitesimals*. This identifies the extra data such a comparison would have to carry; it does not rule one out.

14.6 Comparison of the integration strategies

Strategy	What it supplies	What must not be assumed
Polynomial simplex integration	Finite algebraic Stokes identities at arbitrary field scales	That it defines an integral for every semialgebraic function, or for forms with poles on the chain
Hahn coefficient integration	Exact K -valued periods and Stokes on admitted parameter chains	That fine continuity, or mere pole avoidance, is sufficient
Deformed chains and homotopy	Exact endpoint transport, integer winding, deformed Cauchy formulae	That all contour integrals are infinitesimally deformation invariant; only closed forms are

Strategy	What it supplies	What must not be assumed
Microscopic rescaling	Contour realizations for radius-free analytic-germ residues	That the original coefficient germs share an ordinary radius
Determinant transformation	Coordinate-torus representation of a general isolated complete-intersection residue	That the untransformed integrand has the same torus integral, or that $\det A$ is optional
Algebraic winding and residues	Rational closed-contour pairing at arbitrary scales	A prior Riemann integral, or canonical open-path logarithms
Semialgebraic integration	Finite triangulated chains and Stokes formulations	That every version has the same scalar target
Berkovich–tropical integration	Real differential forms on enlarged analytic spaces	That the original point plane, or all higher valuation ranks, are preserved
Perfectoid cycle integration	An actual chain category with a Stokes formula	That residue characteristic and period-ring target can be changed for free
Surreal antiderivatives	Integration on distinguished extendable function classes	Automatic closure under arbitrary contour substitutions
Roots-of-unity averaging	Exact Laurent extraction; coefficientwise quadrature	That ordinary coefficient limits are valuation limits

15 A usable procedure, and research directions with exact missing steps

15.1 Choose the object before choosing the formula

The most effective first decision is whether the input is a polynomial differential on a nonstandard simplex, a rational function on a semialgebraic loop, a coherent analytic family on an ordinary base, or a radius-free germ. Those choices lead respectively to (5.1), (12.1), (8.2), and a small-scale realization such as [theorems 10.4, 11.1 and 11.4](#). *Merely declaring a function fine-holomorphic selects none of these integrals.* A reliable workflow is the following finite sequence of checks.

1. **Specify the workspace and the category of chain.** Record Γ , the chosen monomials, and whether the parameters are ordinary real parameters, field-valued semialgebraic parameters, or polynomial simplex data. State which notion of interior or component is used, and which of the four rings of [section 2.3](#) the coefficients live in.
2. **Normalize the scale and the integrand.** In a chart $z = a + rw$, compute the pulled-back differential *including* its factor r , as in (7.6). For a meromorphic function, normalize its denominator by a leading monomial and inspect the resulting ordinary leading coefficient on the base contour.
3. **Prove the support certificate.** Identify the source support S and the positive perturbation set E , and check $S + E^*$. Positivity without well-ordering is insufficient; a lower valuation bound without finite contribution at each exponent is also insufficient.
4. **Choose the simplification justified by the hypotheses.** Use Stokes or a coherent primitive for a bounding contour; [theorem 8.5](#) for an evaluation; preparation and residues for finite moving poles; algebraic indices when rational semialgebraic geometry is more informative than a coarse standard part.
5. **Resolve a boundary layer rather than assigning it a side.** When a normalized pole

reduces to the base contour, the coefficient inverse may not exist on a common collar (section 3.6). Move the contour by an explicitly admissible detour, zoom into the offending scale, or use (12.1). State the resulting convention.

6. **Keep equality distinct from convergence.** A strong sum, a coefficientwise limit, a rank-one valuation limit, and a full fine limit are different assertions. Return the exact Hahn result with its support justification, and add an asymptotic or numerical approximation only in an explicitly chosen topology.

At each step record the coefficient algebra, exponent group, parameter domain, differential, and integral target. Those data remove most apparent paradoxes. A proof that uses a derivative-zero fine function, an ordinary real contour, a definable Jordan interior, and a valuation limit without comparison theorems is generally mixing four different constructions.

15.2 A compact decision table

Input	Natural method	Check before applying it
Polynomial differential on a nonstandard simplex or polytope	Factorial simplex integral and theorem 5.1	Finite polynomial presentation; matching oriented faces.
Coherent holomorphic $F \in \mathcal{H}(U)$ on an ordinary base	Hahn contour integral; theorems 8.1 and 8.5	One common coefficient domain and one well-ordered support.
Uniform positive-support contour deformation	Taylor pullback and theorem 8.2	Common support throughout the deformation; nonsingular coefficient collar.
Rational $Q \in K(z)$ on a semialgebraic loop	Algebraic winding weighted by local residues	No pole on the loop; normalization and orientation fixed.
Radius-free one-variable germs in $\mathcal{A}_1$	Rescale to a protecting circle, theorem 10.4	The strict inequality $m\rho < v_H(E)$.
Radius-free coordinate-power system in $\mathcal{A}_n$	Scaled product torus, theorem 11.1	All $d_j\rho_j < v_H(E_j)$.
General isolated complete intersection in $\mathcal{A}_n$	Determinant transformation, theorem 11.4	The integrand must be the <i>transformed</i> one; see section 11.4 .
Root-of-unity sample sequence	Coefficientwise recovery, or a verified valuation-tail criterion	Never infer the topology of the limit from the sampling formula alone.

15.3 Research problems sharpened by these results

A single-cycle realization for arbitrary finite Hahn zero schemes. Given arbitrary isolated f and radius-free positive-support E , construct one admissible n -cycle on which the *untransformed* integrand integrates to (2.4). A solution may require a non-diagonal analytic change of variables, a finite resolution-type decomposition, or chains on an enlarged analytic space, and it must prove that no omitted summation or radius assumption is hidden in the parameterization. The finite-algebra and trace identities (11.9) offer a robust target: any proposed cycle integral should annihilate (F) and satisfy $\text{Res}_F(HJ_F) = \text{Tr}(M_H)$. These are strong compatibility tests; they do not by themselves

construct the cycle. See [section 11.4](#) for exactly how this problem relates to [theorem 11.4](#), which solves the transformed version.

A boundary-layer atlas theorem. Under explicit tame hypotheses, can rational or selected analytic data with poles infinitesimally close to a leading contour be covered by finitely many scale charts whose pullback integrals glue? The values for rational closed contours are already fixed by (12.1); the issue is realizing them by a uniform chain construction that also accommodates open paths and nonrational forms. More generally, an *admissible scale atlas* would need admissible transition pullbacks, a common-refinement property for contour descriptions, and equality of pulled-back integrands on each overlap; [lemma 7.2](#), [theorem 7.4](#), and [proposition 9.1](#) supply local tools, not that global theorem. If local primitives are used, their overlap differences must lie in a controlled constant or period system, or the fine-local logarithm example [example 8.3](#) destroys path invariance.

A comparison theorem with definable integration. Identify a nontrivial common class of functions and chains on which a chosen semialgebraic integration operator, with its actual codomain specified, maps naturally to the Hahn functional. Polynomial simplex forms are a necessary initial test; logarithmic forms and infinitesimal boundary layers are more discriminating, because choices or loss of scale become visible there. Agreement of rational periods is a beginning, not a general theorem.

An intrinsic coherent chain category. The present construction uses explicit ordinary parameterizations and support-controlled pullbacks. A global theory would define admissible atlases and equivalence of chains, prove a gluing theorem, and identify their homology independently of a chosen presentation, addressing possibly unbounded collections of supports rather than assuming that local coherence gives global common-support data. A related concrete question is to compare (6.1) with the larger complex of forms whose supports are uniform only near ordinary base points, especially on noncompact bases and for locally finite chains.

Sectorial transcendental contour theory and higher-rank analytic chains. Essential singularities may require a selected summation or extension operator with compatible monodromy: the formal Laurent expansion of $e^{1/z}$ at a nonzero infinitesimal z has descending valuations and is not strongly summable [1, §13], so fine analyticity alone does not identify an extension or its periods. Separately, the microscopic torus formulas retain every exponent in Γ , and any relationship with analytic skeleta or higher-rank valuations should either preserve those exact periods or state precisely which quotient of the scale information it keeps. Rank-one real-valued geometries and positive-residue-characteristic perfectoid chains cannot be substituted unchanged; the finite separating-torus construction provides test cases and exact invariants for any proposed broader chain theory.

These are research directions formulated by the present analysis, *not* statements that a particular named published conjecture remains open or has been solved.

16 Conclusion

There is no single unqualified Jordan–Cauchy–Stokes theorem on the raw surcomplex point class, and no justified reason to put the three classical statements under one unspecified topology on $\mathbf{No}[z]$. There are, however, precise and useful counterparts. Standard-part geometry restores an exact two-component theorem for a thick boundary at any chosen affine scale. Definable topology restores an exact two-component theorem with definable paths and definable connectedness, together with integer winding. Polynomial simplex integration gives finite algebraic Stokes identities at arbitrary field scales. Hahn differential forms give a surcomplex-valued Stokes pairing with computable cohomology, and support-controlled Taylor pullback extends that pairing to infinitesimally deformed chains, yielding exact endpoint transport, Cauchy’s theorem and formula, winding stability, and

deformed residue contours — all without relying on fine-continuous paths. Positive microscopic rescaling converts radius-free analytic germs into polynomial-coefficient families and gives genuine small-circle and torus residue realizations; residue transformation with its essential determinant represents arbitrary isolated complete-intersection residues by controlled microscopic tori. Rational winding pairings fill an important part of the gap where macroscopic coefficientwise pullbacks fail.

The unifying principle is constructive rather than prohibitive, and it is not to discard infinitesimals or force every scale into one real-valued metric. It is to specify which information each topology retains, to preserve the support algebra, and to prove that boundary, pullback, and integration commute. Within those conditions, nonstandard scales and infinite residues are manageable. Outside them, changing topology or summation method is a new mathematical step that must be stated and justified.

A Support certificates and logical dependencies

The following table records the containing supports used in the proofs. S is a source support, T a second source support, E a positive well-ordered error support, E_a the union of the coordinate supports of a translation point, and $P = \{\rho_1, \dots, \rho_n\}$ a finite positive rescaling alphabet.

Operation	Containing support or finite mechanism
Coefficientwise d , ordinary integration, differentiation, residue extraction, a chosen linear splitting, or the radial homotopy operator (6.4)	S ; no enlargement whatever.
Product of two Hahn families	$S + T$, with finite convolution at each exponent.
Infinitesimal Taylor evaluation (8.1) or deformed pullback (7.3)	$S + E^*$.
Deformed homotopy operator (7.7)	The same $S + E^*$; ordinary integration in the homotopy parameter adds no exponents.
Endpoint correction (8.4)	$S + (\text{supp } \varepsilon)^*$, with at least one displacement factor in every term.
Positive monomial rescaling (10.1)	$S + P^*$; a fixed output exponent has a <i>polynomial</i> coefficient.
Rescaling after infinitesimal translation (10.2)	$S + (E_a \cup P)^*$.
Normalized one-variable denominator, lemma 10.3	Positive error because $v_H(E) - m\rho > 0$; inverse valid on an ordinary base annulus avoiding $w = 0$.
Normalized coordinate-power denominator tuple, (11.2)	Component j has error support $(\text{supp } E_j - d_j \rho_j) + P^*$, positive by the protecting inequality; inverse valid on a common polyannulus.
Perturbation residue after integration, (11.4)–(11.8)	$S + E^*$; before integration the numerator and volume factors may introduce a fixed shift, but a shift of a well-ordered support is still well ordered.
Algebraic rational contour (12.1)	Finite pole set and integer algebraic winding; no infinite sum or limit.
Polynomial simplex integral (5.1)	Finite rational factorial identities in finitely many coefficients.

Operation	Containing support or finite mechanism
Coefficientwise sampling limit (13.5)	A fixed containing support S and ordinary convergence in every coefficient; no valuation-convergence inference.

For clarity, “only finitely many dependencies at an exponent” does not mean “only finitely many exponents below it”; the latter fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$. The proofs use the former, which follows from finite word decompositions. This distinction is essential in lemma 11.3 and in the passage from infinitely many germs to one finite ordinary-parameter calculation.

Several hypotheses have deliberately *not* been weakened. The curve in the topological Jordan theorem may be arbitrary, but integration uses piecewise smooth or polynomial parameterizations. The bounded Jordan domain in the Cauchy formula has closure inside the common coefficient domain. Positive perturbation supports must be well ordered, not just positive. Homotopies share one support on their whole ordinary compact parameter space. Theorem 11.1 assumes coordinate-power leading equations. Rational closed-period integration is a definition, compared to analytic integration only on an explicitly established overlap.

The dependency structure is explicit. Lemma 2.2 defines the Hahn operations. Ordinary Stokes proves theorem 6.1. Taylor pullback and finite-jet identities prove theorem 7.4. Cauchy follows from closedness and ordinary coefficient Cauchy. Independently, positive rescaling and Laurent coefficient extraction realize the perturbation residue, and classical finite-family residue transformation extends the realization to general isolated complete intersections. The rational pairing is defined independently from semialgebraic winding and compared with Hahn integration only afterward. *No Jordan statement in the fine topology, and no unproved general fine integration rule, enters this chain of arguments.* The new proofs do not depend on the Noetherian or Nullstellensatz claims of [2] except where that manuscript’s finite-zero and residue results are expressly invoked — in (11.9) and, conditionally and with that dependence stated, in theorem 11.5. In particular theorems 5.1, 6.1, 6.3, 7.5 and 11.4 and the sampling comparisons stand independently of those deeper analytic-algebra claims.

B Why finite-parameter testing proves residue transformation

This appendix supplies the technical point used in lemma 11.3 and in theorem 11.5, without assuming an infinite-parameter contour for the original germ family. It is stated for a transformation matrix C arising from a linear splitting, which is the harder case; the ordinary analytic matrix A of (11.6) is a special case in which the convergence discussion of section B.2 is unnecessary.

B.1 Finite dependency

In the supplied normal-form construction an ordinary linear splitting for $\mathbb{C}\{z\}/(f)$ is extended coefficientwise, and the perturbed inverse is a Neumann series in operators carrying positive perturbation support. Therefore each coefficient of M_i , of Q_i , and of the chosen C_{ij} is built from a finite sum of positive-support words. The residue expansion (2.4) has the same property: for a numerator with support S , its coefficient at λ uses only decompositions

$$\lambda = \sigma + e_1 + \cdots + e_q, \quad \sigma \in S, \quad e_j \in E,$$

of which there are finitely many, including finitely many possible lengths, by [lemma 2.2](#). *This does not say that only finitely many exponents lie below λ .*

Fix a coefficient to be checked in the transformation identity. Retain all ordinary source germs and perturbation germs occurring in its finite dependency sets on both sides, treating numerator source coefficients separately by linearity. Replace the finitely many retained positive perturbation monomials by independent parameters $u_1, \dots, u_b$. The normal-form formulae now define formal power series in u with ordinary analytic germ coefficients, and their formal identities include $Q(u) = C(u)F(u)$ with $Q_i(0, z) = z_i^\mu$. All omitted input terms are irrelevant to the selected coefficient by construction.

B.2 Why a formal matrix causes no convergence gap

The auxiliary ordinary linear splitting need not be continuous in an analytic norm, so $C(u)$ cannot be asserted convergent as an analytic parameter series. Let N exceed every total parameter degree needed for the selected coefficient and let $C_{\leq N}(u)$ be its finite truncation; its finitely many coefficient germs *do* have a common ordinary representative neighborhood. Define the actual finite analytic family

$$\tilde{Q}(u, z) = C_{\leq N}(u, z) F(u, z),$$

which has ordinary leading tuple $(z_1^\mu, \dots, z_n^\mu)$ and agrees with the formal $Q(u, z)$ modulo $(u)^{N+1}$. The tuple $F(u, z)$ has leading isolated complete intersection f , so after choosing ordinary small representatives both tuples have finite clusters near zero.

The ordinary residue transformation rule, applied to these finite families and their clusters, gives

$$\text{res}_{F(u)}^{\text{cluster}}(H) = \text{res}_{\tilde{Q}(u)}^{\text{cluster}}(H \det C_{\leq N}(u)),$$

the family form of classical residue transformation, which follows from the local transformation law and additivity over the finitely many points [\[9, 28\]](#). Expanding the reciprocal denominators on ordinary protecting residue cycles identifies their Taylor series with the geometric residue expansions, and a Taylor coefficient through total parameter degree N uses only denominator and numerator coefficients through that degree. Consequently the last display proves the formal transformation law through degree N for the original Q and C .

Finally substitute $u_j = t^{e_j}$ and extract the chosen Hahn coefficient. Several parameter monomials may contribute to the same exponent, but only finitely many do, and their finite sum gives the desired coefficient equality. Repeating the argument at each exponent establishes the transformation identity. The auxiliary ordinary cycles may change from exponent to exponent; *the final microscopic separating torus is constructed independently*, by [theorem 10.1](#) and the radius inequalities [\(11.2\)](#) or [\(11.13\)](#).

C What is claimed, what is imported, and what is not proved

C.1 Consolidated status statement

Established results, statements reported in the two supplied manuscripts, and deductions proved here are distinguished throughout. The two supplied manuscripts [\[1, 2\]](#) are treated as research manuscripts, not as independently refereed references; they are not modified or redistributed, and their source filenames are recorded in the bibliography so that the direction of dependence can

be checked. The literature search was conducted on September 21, 2026 and was targeted rather than exhaustive. *This is not a claim of a unique accepted foundation for surcomplex analysis, an exhaustive priority or novelty determination, or a formally verified proof.* No named published conjecture is claimed to be solved. Symbolic checks validate the displayed finite examples — exact expressions, Taylor coefficients, matrix identities, residue traces, finite roots-of-unity averages, polynomial Stokes on monomial forms, and endpoint transport on polynomial deformations — and are *not* machine verification of the general theorems: they do not implement arbitrary Hahn fields, enumerate arbitrary ordered supports, decide arbitrary surreal comparisons, or prove semialgebraic Jordan separation. All sums and computational workspaces are set-sized; no proper-class summation is used. The priority position is that the value of this package is its complete set of hypotheses; *no priority claim is based merely on not finding an identical formulation in the literature*, and the constructions proved here are proposed extensions of the supplied presentation, not a claim that their underlying coefficientwise or residue-theoretic ideas are absent from the literature. This article is not an independent audit of every theorem in either supplied manuscript.

C.2 Per-theorem non-claims, preserved

The following specific non-claims are part of the results they accompany and are repeated here for reference.

- *On the sources.* Neither supplied manuscript establishes a Jordan theorem for arbitrary fine-topological curves, nor generalized Stokes for arbitrary fine-smooth functions and arbitrary surcomplex domains. We do not silently attribute either assertion to them.
- *On [proposition 3.1](#).* This is not a theorem that a Hahn field with its own valuation topology is discrete; the radius supplied by the cut may lie outside the workspace.
- *On [remark 3.3](#).* A claim about convergence or connectedness in one of the three topologies is not automatically a claim in either of the others.
- *On [theorem 4.1](#).* The separating object is the thickened curve, not the ordinary point set; neither component is called fine-connected; two infinitesimally separated points in one boundary monad cannot be assigned different sides, and the theorem cannot distinguish two zeros lying in one standard-part fibre; an unbounded exterior in the internal ordered-field sense has not been obtained.
- *On [theorem 4.3](#).* This is an application of semialgebraic topology, not a new surreal theorem; the technical finite-topology input is imported, not replaced by an informal appeal to transfer of arbitrary continuous maps; the components are not connected in the unrestricted intrinsic topology; the o-minimal generalization refers to that framework’s differentiability and definability hypotheses, not to arbitrary class functions or arbitrary fine-topological subsets; definable results do not supply ordinary singular homology of a proper class.
- *On [winding](#).* The definable degree is an ordinary integer, not a surreal number of turns; it does not by itself define the integral of a general differential form; the primitive of a semialgebraic function need not be semialgebraic; the availability of a finite algebraic prescription does not make arbitrary surreal inputs computable.
- *On [proposition 3.5](#).* A polynomial antiderivative still gives $\int_0^1 x dx = 1/2$; the obstruction is to a particular unrestricted finite-mesh definition, not to all integration. Hyperfinite partitions require an additional nonstandard universe and transfer structure, supplied neither by the ordered-field notation nor by the surreal normal form.
- *On [theorem 5.1](#).* The proof uses classical Stokes to certify a finite algebraic identity; it does not transfer a statement quantifying over arbitrary integrable functions. It does not define an

integral for every semialgebraic function, nor for a rational form whose denominator vanishes on the chain.

- On [theorem 6.3](#). For infinite-dimensional ordinary cohomology, $V((t^\Gamma))$ must not be silently replaced by $K \otimes_{\mathbb{C}} V$; and the global uniform-support assignment is not claimed to be a sheaf for arbitrary infinite covers.
- On [definition 7.1](#) and [lemma 7.2](#). Φ describes an admissible parameterized chain, not a fine-continuous map from the ordinary parameter manifold; the jet prolongation is the *specified* extension, with no claim of ordinary Taylor convergence or of a canonical prolongation of every smooth function on every fine domain.
- On [theorem 7.5](#). It is incorrect to claim that all contour integrals are infinitesimally deformation invariant; closedness is the controlling condition.
- On [proposition 7.6](#). Ordered estimates do not imply convergence of set-indexed nets in the full fine topology.
- On [theorem 10.1](#). The conclusion is a new coherent representation in an infinitesimal chart, not ordinary convergence of the original germs on a uniform disk; it neither establishes nor requires all-scale coherence.
- On [theorem 10.5](#). An arbitrary Hahn family of meromorphic germs is larger than a quotient of two elements of $\mathcal{A}_1$, and is not claimed to admit the microscopic contour construction.
- On [theorem 11.4](#). The untransformed form is not claimed to have the same torus integral; $\det A$ is not an optional Jacobian factor, since AF can introduce extra zeros; and F and G are not claimed to have identical zero schemes. [Section 11.4](#) records that one of the merged sources judged the general realization unproved, and that the alternative route [theorem 11.5](#) is conditional on [2].
- On [remark 11.6](#). Formula (11.14) does not assert regularity of the original quotient on the chosen torus; omitting the determinant is generally wrong; a real n -torus is not identified with the full boundary of a complex polydisk.
- On [definition 12.1](#). Equation (12.1) is a definition, not a claim that a finite-partition Riemann net converges to this value; it does not determine open-path logarithms and is not a definition on all of $\mathcal{H}(U)$ or $\mathcal{A}_n$.
- On [theorem 13.2](#). Coefficientwise recovery does not make the Hahn integral a fine limit, and in general does not make it a valuation limit either.

D Notation

Symbol	Meaning
$\mathbf{SC} = \mathbf{No}[i]$	The full surcomplex class field.
Γ, R_Γ, K	A set-sized divisible ordered subgroup of $\mathbf{No}$, and the real and complex Hahn workspaces $\mathbb{R}((t^\Gamma))$ and $\mathbb{C}((t^\Gamma))$, with $t^\gamma = \omega^{-\gamma}$.
v, v_H	Least exponent of a field element; least exponent of a Hahn coefficient-family datum.
$K^\circ, \mathfrak{m}_K, \mathfrak{m}, \text{st}$	Finite elements; infinitesimals of the workspace; the class of all infinitesimals of $\mathbf{SC}$; the standard-part (reduction) map $K^\circ \rightarrow \mathbb{C}$.
$U^\sharp$	$\text{st}^{-1}(U) = \{c + \varepsilon : c \in U, \varepsilon \in \mathfrak{m}_K\}$.
Standard-part topology	The initial topology of st ; called the residue, reduction, or shadow topology elsewhere.
E^*	The additive monoid of finite words in a well-ordered positive support E , including 0.

Symbol	Meaning
$\mathcal{H}(U)$	$\mathcal{O}(U)((t^\Gamma))$: one common ordinary coefficient domain.
$\mathcal{A}_n$	$\mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$: radius-free coefficient germs.
$\mathbb{C}[w]((t^\Gamma))$	Polynomial-coefficient ring; the target of theorem 10.1 .
$\Omega_H^q(M)$	$\Omega^q(M; \mathbb{C})((t^\Gamma))$: globally supported Hahn series of ordinary smooth complex q -forms.
$\Phi = \phi_0 + \eta$	Ordinary base parameterization plus a uniformly supported infinitesimal deformation, definition 7.1 .
$\int^H, \oint^H$	Hahn coefficientwise integration and its closed-contour form.
$\int^{\text{pol}}, I_{\text{alg}}$	Polynomial-simplex integration (5.1); the algebraic rational contour pairing (12.1).
$T_F(c, \varepsilon)$	Microscopic endpoint Taylor correction (8.4).
wind, wind _{sa}	Ordinary complex winding; definable/algebraic integer winding.
res _a , Res _F	Ordinary local residue (Laurent or Grothendieck); the strong Hahn perturbation residue (2.4).
m, μ, e_a	Order of a one-variable germ; global colength; local multiplicity at an actual displaced zero, $\sum_a e_a = \mu$.
$\rho_j, r_j = t^{\rho_j}$	Rescaling exponents and the corresponding microscopic radii.
d_j	Coordinate powers in a coordinate-power leading system (11.1).
J_F, M_H, B_F	Jacobian determinant; multiplication operator; the finite quotient algebra $\mathcal{A}_n/(F)$.

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